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检测
TESTING
CNAS L1020

实验室名称：国家电器产品质量监督检验中心
Lab Name: China National Center for Quality Supervision and Test of Electrical Apparatus Products

№ 21M2079-S

检验(试验)报告
Test Report

委 托 单 位：Fuzhou Tianyu Electric Co., Ltd.
Client:

产 品 名 称：Power transformer
Name of Product:

产 品 型 号：SFZ-150000/132
Product Type:

检 验 类 别：Commission test
Test Category:

本实验室对出具的检验(试验)结果负责，未经实验室书面同意，不得部分地复制本报告.

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China National Center for Quality Supervision and Test of Electrical Apparatus Products

Test Report

№: 21M2079-S

Client	Fuzhou Tianyu Electric Co., Ltd.	Test category	Commission test
Manufacturer	Fuzhou Tianyu Electric Co., Ltd.	Date of sample receiving	/
Name of sample	Power transformer	Type of sample	SFZ-150000/132
Address of manufacturer	No. 28, Yaoxi Road, Nanyu Town, Minhou County, Fuzhou City, Fujian Province	Original number or date of production	AN17901
Date of test	From Aug. 11, 2021 to Aug. 19, 2021	Number of sample	1 set
Test items	Routine test Type test (including calculation of the winding hot-spot temperature-rise) Measurement of bushing capacitances and dielectric dissipation factor (tanδ) Measurement of frequency response Measurement of no-load excitation characteristics Long-duration no-load test Measurement of zero-sequence impedances on three-phase transformers Measurement of the harmonics of the no-load current	Test standards	GB/T 1094.1—2013 GB/T 1094.2—2013 GB/T 1094.3—2017 GB/T 1094.10—2003 GB/T 6451—2015 GB/T 7595—2017 JB/T 10088—2016 IEC 60076-1: 2011 IEC 60076-2: 2011 IEC 60076-3: 2013+AMD1: 2018 IEC 60076-10: 2016 Commission requirements
Test conclusion	The test results of routine test, type test (including calculation of the winding hot-spot temperature-rise), measurement of bushing capacitances and dielectric dissipation factor (tanδ), measurement of frequency response, measurement of no-load excitation characteristics, long-duration no-load test, measurement of zero-sequence impedances on three-phase transformers and measurement of the harmonics of the no-load current of power transformer (type: SFZ-150000/132) are in accordance with test standards and commission requirements. The sample has passed the above tests. 国家电器产品质量监督检验中心 检验检测专用章 (6) Signing and issuing date: 2021.09.14 Note: the conclusion is valid only for the inspected and tested sample.		
Remarks	All tests were performed in Fuzhou Tianyu Electric Co., Ltd.		

Compiled by: 于浩 Proofread by: [signature] Checked by: 王占宾 Approved by: [signature]

1. Sample parameters

Rated power: 150000kVA

Rated voltage: 132/21kV

Rated current: 656.1/4124A

Rated frequency: 50Hz

Number of phases: 3

Tapping ranges: ±9×1.67%

Connection symbol: YNd11

Cooling method: ONAN/ONAF (120000/150000kVA)

Class of insulation and heat-resistant: A

Insulation level:	HV	Um/LI/LIC/AC	145/650/715/275kV
	HVN	Um/LI/AC	72.5/325/140kV
	LV	Um/LI/LIC/AC	24/125/140/55kV

2. Test standards

GB/T 1094.1—2013 *Power transformers—Part 1: General*

GB/T 1094.2—2013 *Power transformers—Part 2: Temperature rise for liquid-immersed transformers*

GB/T 1094.3—2017 *Power transformers—Part 3: Insulation levels, dielectric tests and external clearances in air*

GB/T 1094.10—2003 *Power transformers—Part 10: Determination of sound levels*

GB/T 6451—2015 *Technical parameters and requirements of oil-immersed power transformer*

GB/T 7595—2017 *Quality of transformer oils in service*

IEC 60076-1: 2011 *Power transformers—Part 1: General*

IEC 60076-2: 2011 *Power transformers—Part 2: Temperature rise for liquid-immersed transformers*

IEC 60076-3: 2013+AMD1: 2018 *Power transformers—Part 3: Insulation levels, dielectric tests and external clearances in air*

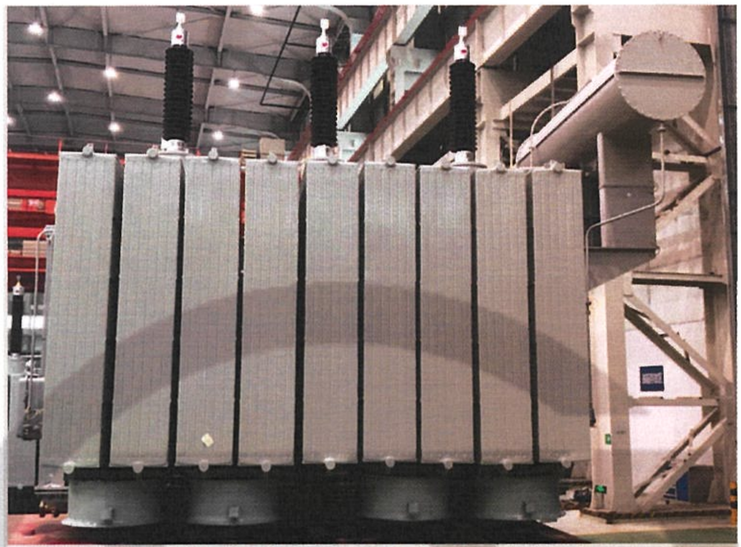
IEC 60076-10: 2016 *Power transformers—Part 10: Determination of sound levels*

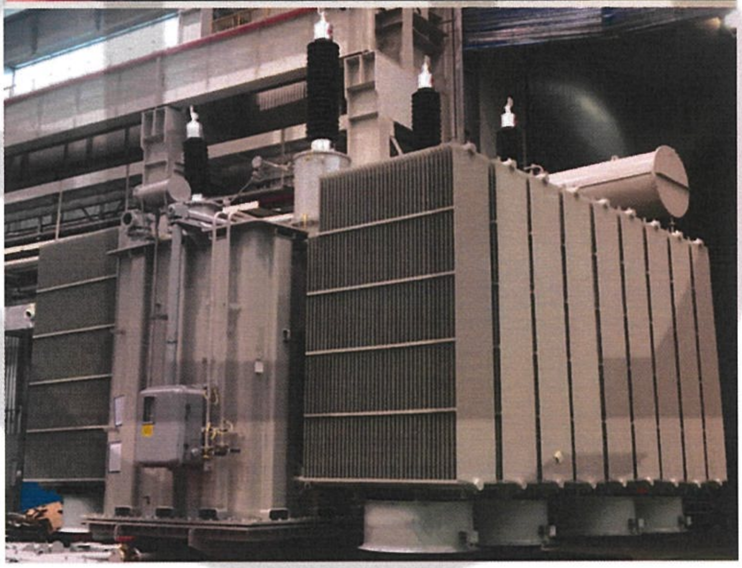
Commission requirements

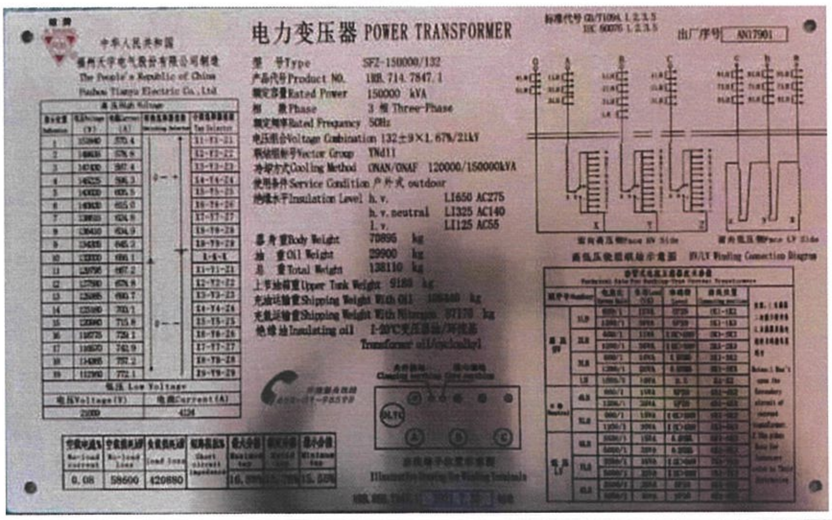
3. Sample description

The power transformer is for outdoor use, external photos of the sample have been attached.

Photos of the sample







Summary of test results

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
1	Measurement of d.c. insulation resistance between each winding to earth and between windings (routine test)	Providing value of insulation resistance Providing absorption ratio Providing polarity index	See 4.1	/
2	Check of core and frame insulation for liquid-immersed transformers with core or frame insulation (routine test)	Providing value of insulation resistance at 20°C(GΩ): ≥1.0	See 4.2	PASS
3	Measurement of dissipation factor (tanδ)of the insulation system capacitances (routine test)	Providing dielectric dissipation factor tanδ at 20°C: tanδ: ≤0.8%	See 4.3	PASS
4	Determination of capacitances windings-to-earth and between windings (routine test)	Providing value of capacitance	See 4.4	/
5	Measurement of bushing capacitances and dielectric dissipation factor (tanδ) (commission test)	Providing value of capacitance Providing dielectric dissipation factor tanδ	See 4.5	/
6	Check of the ratio and polarity of built-in current transformers (routine test)	Providing the value of ratio and polarity relation	See 4.6	/
7	Insulation test of auxiliary wiring (AuxW) (routine test)	Wiring for auxiliary power and control circuit: 2.0kV 60s Wiring for secondary winding of current transformer: 2.5kV 60s	2.0kV 60s 2.5 kV 60s	PASS
8	Measurement of voltage ratio and check of phase displacement (routine test)	Voltage ratio tolerance of principal tapping: obtaining the lower of the following values between ±0.5% of declared ratio and ±1/10 of the actual percentage impedance Connection symbol: YNd11	See 4.8	PASS
9	Measurement of winding resistance (routine test)	Maximum resistance unbalance rate Phase resistance: ≤2% Line resistance: ≤1%	HV(phase): 0.51% LV(line): 0.28%	PASS

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
10	Measurement of no-load loss and current (routine test)	I_0(%): 0.30 +0% P_0(kW): 69.6 +0%	0.07 58.500	PASS
11	Measurement of no-load loss and current at 90% and 110% of rated voltage (routine test)	I_0(%): measured P_0(kW): measured	90% / 110% 0.04 / 0.75 42.034 / 99.280	/
12	Measurement of short-circuit impedance and load loss (routine test)	ONAF: t: 75°C Z(%): 15.5 ±3% P_k(kW): 422.000 +0% P_{total} (kW): 491.600 +0%	 15.76 420.6614 479.1610	PASS
		ONAN: t: 75°C Z(%): measured P_k(kW): measured P_{total}(kW): measured	 12.61 269.223 327.723	
13	Lightning impulse test (LI, LIC, LIN) (routine test, type test)	HV: Full wave (kV): 650 ±3% Chopped wave (kV): 715 ±3% Neutral point (kV): 325 ±3%	 636.08～656.74 712.27～722.42 325.47～327.26	PASS
		LV: Full wave (kV): 125 ±3% Chopped wave (kV): 140 ±3%	 124.33～127.52 136.25～142.67	
14	Tests on on-load tap-changers (routine test)	According to clause 11.7 of GB/T 1094.1-2013	Meet the requirements	PASS
15	Applied voltage test (AV) (routine test)	HV neutral point: 140kV 60s LV: 55kV 60s	140.0kV 60s 55.0kV 60s	PASS
16	Line terminal AC withstand test (LTAC) (routine test)	Test on phase to earth Applied voltage (kV): Induced voltage (kV): 275 Duration (s): 120(f_n/f) Frequency (Hz): ＞50	 ac: 43.92 / ab: 43.92 / bc: 43.92 A: 275.0 / B: 275.0 / C: 275.0 30 200	PASS

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
17	Induced voltage withstand test and induced voltage test with partial discharge measurement (IVW, IVPD) (routine test)	Three-phase applied voltage		PASS
		$0.4Ur/\sqrt{3}$ (kV) Discharge magnitude ≤50pC	30.5 A:<20; B:<20; C:<20	
		$1.2Ur/\sqrt{3}$ (kV) Duration (min): 1 Discharge magnitude ≤100pC	91.5 1 A:<40; B:<40; C:<40	
		$1.58Ur/\sqrt{3}$ (kV) Duration (min): 5 Discharge magnitude ≤100pC	120.4 5 A:<60; B:<60; C:<60	
		$2.0Ur/\sqrt{3}$ (kV) Duration (min): 0.5	152.4 0.5	
		$1.58Ur/\sqrt{3}$ (kV) Duration (min): 60 Discharge magnitude ≤100pC	120.4 60 A:<60; B:<60; C:<60	
		$1.2Ur/\sqrt{3}$ (kV) Duration (min): 5 Discharge magnitude ≤100pC	91.5 5 A:<50; B:<50; C:<50	
		$0.4Ur/\sqrt{3}$ (kV) Duration (min): 1 Discharge magnitude ≤50pC	30.5 1 A:<20; B:<20; C:<20	
		Frequency (Hz): >50	200	
18	Insulating liquid test, measurement of dissolved gasses in dielectric liquid from each separate oil compartment except diverter switch compartment (routine test, type test, commission test)	Breakdown voltage (kV): ≥60 tanδ(90°C): ≤0.3% Water content (mg/L): ≤10 Flash point (closed cup) (°C): >170	61.8 0.11% 7.3 175	PASS
		Providing gas chromatograph analysis: Hydrogen: <10μL/L Acetylene: 0 Total hydrocarbon: <10μL/L	See 4.18	
19	Measurement of frequency response (special test)	Providing frequency response characteristics curve	See 4.19	PASS
20	Measurement of no-load excitation characteristics (commission test)	Providing no-load excitation characteristic curve	See 4.20	/
21	Long-duration no-load test (commission test)	Applied voltage (kV): 1.1Ur Duration (h): 12 Without acetylene in oil	1.1Ur 12 For gas chromatograph analysis, see 4.21	PASS
22	Measurement of zero-sequence impedances on three-phase transformers (special test)	Providing zero-sequence impedances values (Ω)	See 4.22	/
23	Measurement of the harmonics of the no-load current (commission test)	Providing harmonics of the no-load current of each phase	Harmonics of the no-load current of I_1-I_{19}	/

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
24	Temperature-rise test (including calculation of the winding hot-spot temperature-rise) (type test)	Cooling method: ONAN Top oil temperature-rise limits (K): 53 Winding temperature-rise limits (K): 60 Winding hot-spot temperature-rise limits (K): 73 Temperature-rise of tank and structural parts (K): 70	Top oil temperature-rise: 49.7 HV winding temperature-rise: 57.0 LV winding temperature-rise: 57.0 HV winding hot-spot temperature-rise: 72.0 LV winding hot-spot temperature-rise: 71.8 Temperature-rise limits of tank and structural parts: 54.7	PASS
		Cooling method: ONAF Top oil temperature-rise limits (K): 53 Winding temperature-rise limits (K): 60 Winding hot-spot temperature-rise limits (K): 73 Temperature-rise of tank and structural parts (K): 70	Top oil temperature-rise: 50.2 HV winding temperature-rise: 55.4 LV winding temperature-rise: 55.8 HV winding hot-spot temperature-rise: 71.8 LV winding hot-spot temperature-rise: 72.0 Temperature-rise limits of tank and structural parts: 55.3	PASS
25	Leak testing with pressure for liquid-immersed transformers (routine test)	Body: Applied pressure (kPa): 30 Duration (h): 24 No oil leakage or damage	30.0 24 No oil leakage or damage	PASS
		On-load tap-changer tank: Applied pressure (kPa): 30 Duration (h): 24 No oil leakage or damage	30.0 24 No oil leakage or damage	
26	Determination of sound levels (type test)	Cooling method: ONAN Sound pressure level $\overline{L_{PA}}$ dB(A): ≤60 Sound power level $L_{WA,\ UN}$dB(A): ≤90	58 80	PASS
		Cooling method: ONAF Sound pressure level $\overline{L_{PA}}$ dB(A): ≤60 Sound power level $L_{WA,\ UN}$dB(A): ≤90	59 82	
27	Mechanical strength test of tank (type test)	Applied vacuum degree (kPa): 0.133 Applied positive pressure (kPa): 100 Elastic deformation of tank wall (mm): ≤30 Elastic deformation of tank cover (mm): ≤37.5 Permanent deformation of tank wall (mm): ≤15 Permanent deformation of tank cover (mm): ≤12.5 No damage	See 4.27	PASS

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4. Test items and results

4.1 Measurement of d.c. insulation resistance between each winding to earth and between windings (routine test)

Test date: Aug. 16, 2021
Relative humidity: 55%; Oil temperature: 33.0°C

Measured parts	R_{15} (GΩ)	R_{60} (GΩ)	R_{600} (GΩ)	Absorption ratio (R_{60}/R_{15})	Polarity index (R_{600}/R_{60})
HV—LV and earth	4.76	7.00	14.10	1.47	2.01
LV—HV and earth	7.40	10.40	20.40	1.40	1.96
HV, LV—earth	6.27	8.52	15.10	1.35	1.77
HV—LV	5.62	7.65	15.60	1.36	2.04

4.2 Check of core and frame insulation for liquid-immersed transformers with core or frame insulation (routine test)

Test date: Aug. 16, 2021
Relative humidity: 55%; Oil temperature: 33.0°C

Measured parts	Measured insulation resistance (GΩ)	Corrected insulation resistance at 20°C (GΩ)
Core—earth	2.46	4.17
Frame—earth	2.35	3.98

4.3 Measurement of dissipation factor (tanδ) of the insulation system capacitances (routine test)

Test date: Aug. 16, 2021
Relative humidity: 55%; Oil temperature: 33.0°C

Measured parts	Measured dielectric dissipation factor tanδ(%)	Dielectric dissipation factor tanδ corrected to 20°C (%)	Capacitance (pF)
HV—LV and earth	0.22	0.12	11180
LV—HV and earth	0.25	0.18	20130
HV, LV—earth	0.26	0.19	18180
HV—LV	0.22	0.16	9260

4.4 Determination of capacitances windings-to-earth and between windings (routine test)

Test date: Aug. 16, 2021

See 4.3.

4.5 Measurement of bushing capacitances and dielectric dissipation factor (tanδ) (commission test)

Test date: Aug. 16, 2021
Relative humidity: 55%; Oil temperature: 33.0°C

Measured contents	Applied voltage	Test parts			
		A	B	C	O
		2101507	2101512	2101508	2123099
tanδ(%)	10kV	0.28	0.29	0.28	0.28
Measured capacitance (pF)		384.3	397.5	392.3	392.5

4.6 Check of the ratio and polarity of built-in current transformers (routine test)

Test date: Aug. 16, 2021

Installation position	Terminal	Standard ratio	Measured ratio	Polarity relation
A	1K1-1K2	600/1	604.1	-
	1K1-1K3	1200/1	1199.4	-
	2K1-2K2	600/1	600.9	-
	2K1-2K3	1200/1	1199.9	-
	3K1-3K2	600/1	600.2	-
	3K1-3K3	1200/1	1199.1	-
B	1K1-1K2	600/1	600.2	-
	1K1-1K3	1200/1	1199.2	-
	2K1-2K2	600/1	600.2	-
	2K1-2K3	1200/1	1199.6	-
	3K1-3K2	600/1	600.3	-
	3K1-3K3	1200/1	1198.5	-
	K1-K2	1000/1	999.8	-
C	1K1-1K2	600/1	601.0	-
	1K1-1K3	1200/1	1199.3	-
	2K1-2K2	600/1	599.9	-
	2K1-2K3	1200/1	1199.6	-
	3K1-3K2	600/1	599.9	-
	3K1-3K3	1200/1	1199.9	-
O	4K1-4K2	600/1	599.9	-
	4K1-4K3	1200/1	1199.7	-
	5K1-5K2	600/1	599.5	-
	5K1-5K3	1200/1	1199.3	-

4.7 Insulation test of auxiliary wiring (AuxW) (routine test)

Test date: Aug. 16, 2021

Relative humidity: 55% ; Ambient temperature: 32.5°C; Air pressure: 102kPa

Parts of applied voltage		Test voltage (kV)	Test duration (s)	Result
Wiring for auxiliary power and control circuit	Powering for on-load tap-changer	2.0	60	PASS
Wiring for secondary winding of current transformer	A	2.5	60	
	B	2.5	60	
	C	2.5	60	
	O	2.5	60	
	a	2.5	60	
	b	2.5	60	
	c	2.5	60	

4.8 Measurement of voltage ratio and check of phase displacement (routine test)

Test date: Aug. 16, 2021

HV winding		LV winding		Transformer ratio by calculation	Measured voltage ratio tolerance (%)			Connection symbol
Tapping position	Voltage (kV)	Tapping position	Voltage (kV)		AB/ab	BC/bc	CA/ca	
1	151.840	/	21	7.230	-0.11	-0.10	-0.10	YNd11
2	149.635			7.125	-0.13	-0.13	-0.13	
3	147.430			7.020	-0.17	-0.15	-0.15	
4	145.225			6.915	-0.18	-0.18	-0.18	
5	143.020			6.810	-0.20	-0.20	-0.20	
6	140.820			6.706	-0.25	-0.22	-0.22	
7	138.615			6.601	-0.27	-0.27	-0.27	
8	136.410			6.496	-0.06	-0.06	-0.06	
9	134.205			6.391	-0.09	-0.08	-0.08	
10	132.000			6.286	0.13	0.14	0.14	
11	129.795			6.181	0.11	0.11	0.11	
12	127.590			6.076	0.09	0.09	0.09	
13	125.385			5.971	0.06	0.06	0.06	
14	123.180			5.867	0.03	0.04	0.04	
15	120.980			5.761	0.01	0.01	0.02	
16	118.775			5.656	-0.03	-0.01	-0.01	
17	116.570			5.551	-0.05	-0.04	-0.03	
18	114.365			5.446	0.21	0.22	0.22	
19	112.160			5.341	0.19	0.19	0.19	

4.9 Measurement of winding resistance (routine test)

Test date: Aug. 16, 2021
Oil temperature: 33.0°C

Winding	Tapping position	Measured values (Ω)			Resistance unbalance rate (%)
		A～O a～b	B～O b～c	C～O c～a	
HV	1	0.1488	0.1489	0.1490	0.07
	2	0.1458	0.1464	0.1459	0.37
	3	0.1427	0.1430	0.1428	0.22
	4	0.1399	0.1399	0.1400	0.15
	5	0.1370	0.1367	0.1372	0.16
	6	0.1339	0.1340	0.1342	0.16
	7	0.1310	0.1310	0.1312	0.08
	8	0.1284	0.1285	0.1286	0.24
	9	0.1255	0.1256	0.1257	0.23
	10	0.1225	0.1226	0.1225	0.08
	11	0.1257	0.1258	0.1260	0.15
	12	0.1289	0.1288	0.1291	0.51
	13	0.1318	0.1319	0.1319	0.57
	14	0.1347	0.1349	0.1349	0.14
	15	0.1377	0.1371	0.1378	0.07
	16	0.1406	0.1399	0.1407	0.13
	17	0.1436	0.1436	0.1438	0.07
	18	0.1462	0.1462	0.1463	0.37
	19	0.1490	0.1492	0.1492	0.22
LV	/	6.537×10^{-3}	6.523×10^{-3}	6.541×10^{-3}	0.28

4.10 Measurement of no-load loss and current (routine test)

Test date: Aug. 16, 2021

Voltage multiple	Applied voltage (kV)		No-load current		No-load loss (kW)	
	Average value	r.m.s. value	(A)	(%)	Measured value	Corrected value
90%Ur	18.928	18.928	1.61	0.04	42.034	42.034
100%Ur	21.005	21.037	2.79	0.07	58.590	58.500
110%Ur	23.132	23.798	30.93	0.75	102.222	99.280

Remarks: at 100%Ur, the difference between r.m.s voltage and average voltage is within 3%.

4.11 Measurement of no-load loss and current at 90% and 110% of rated voltage (routine test)
See 4.10.

4.12 Measurement of short-circuit impedance and load loss (routine test)

Test date: Aug. 16, 2021

Oil temperature: 33.0°C

ONAF:

Winding	Tapping position	Applied current I		Measured voltage (kV)	Short-circuit impedance (for each phase)		Load loss (kW)	Total loss (kW)
		(A)	I/Ir (%)		HV impedance (Ω)	(%)	Corrected value	Corrected value
					t=75°C I=Ir	t=75°C I=Ir	t=75°C I=Ir	t=75°C I=Ir
HV-LV	1	298.74	52.38	13420	25.94	16.88	416.044	474.544
	10	349.62	53.29	11085	18.31	15.76	420.661	479.161
	19	390.21	50.54	8815	13.04	15.55	540.240	598.740

ONAN:

Winding	Tapping position	Applied current I		Measured voltage (kV)	Short-circuit impedance (for each phase)		Load loss (kW)	Total loss (kW)
		(A)	I/Ir (%)		HV impedance (Ω)	(%)	Corrected value	Corrected value
					t=75°C I=Ir	t=75°C I=Ir	t=75°C I=Ir	t=75°C I=Ir
HV-LV	1	298.74	65.47	13420	25.94	12.50	266.250	324.750
	10	349.62	66.61	11085	18.31	12.61	269.223	327.723
	19	390.21	63.17	8815	13.04	12.44	345.752	404.252

4.13 Lightning impulse test (LI, LIC, LIN) (routine test, type test)

Test date: Aug. 16, 2021

Atmospheric conditions of test:

Relative humidity: 55%; Ambient temperature: 32.5°C; Oil temperature: 33.0°C; Air pressure: 102kPa.

Test items and voltage

Withstand terminals	Rated withstand voltage (kV)		Tapping position
	Lightning full wave	Lightning chopped wave	
A, B, C	650	715	A:1; B:10; C:19
O	325	/	1
a, b, c	125	140	/

Test sequence:

Line terminal:

One negative reduced level full impulse;

One negative rated level full impulse;

One negative reduced level chopped impulse;

Two negative rated level chopped impulse;

Two negative rated level full impulse.

Neutral point :

One negative reduced level full impulse;

Three negative rated level full impulse.

Test records:

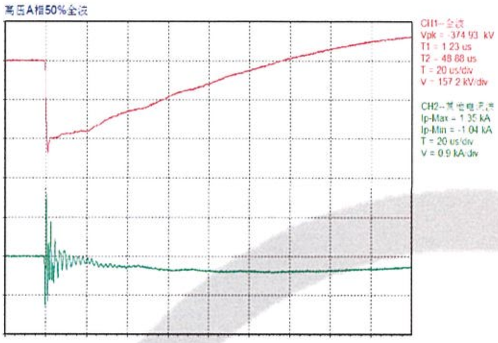
T1: wave front time; T2: time to half peak value; Tc: chopped time; Upk: peak voltage.

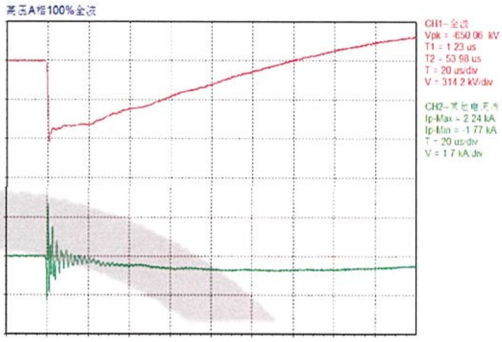
For waveform diagram, see $P_{15\sim21}$.

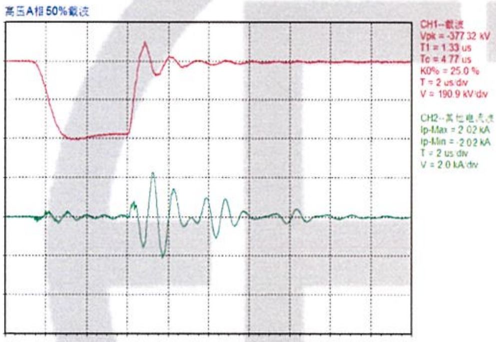
Voltage ranges of oscillograms are as below:

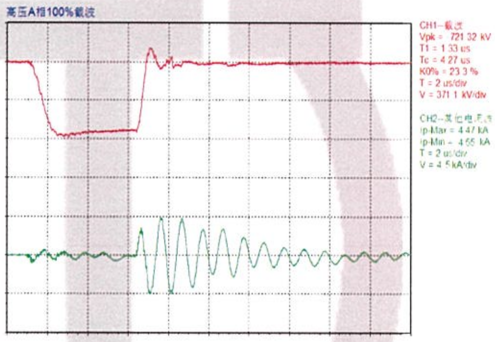
Withstand terminals	Full wave (kV)	Chopped wave (kV)
A, B, C	636.08～656.74	712.27～722.42
O	325.47～327.26	/
a, b, c	124.33～127.52	136.25～142.67

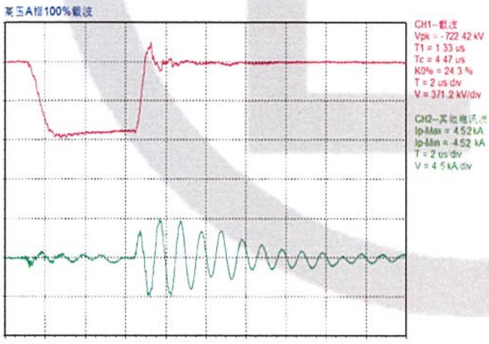
Tested terminal: A Test polarity: negative Channel 1: voltage wave Channel 2: current wave

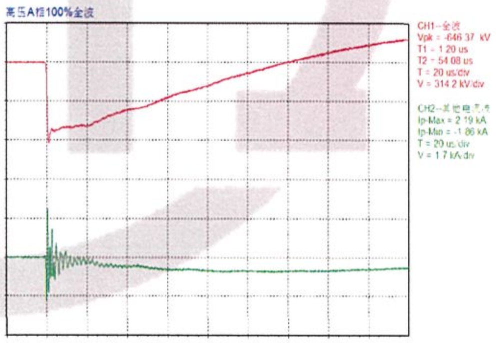


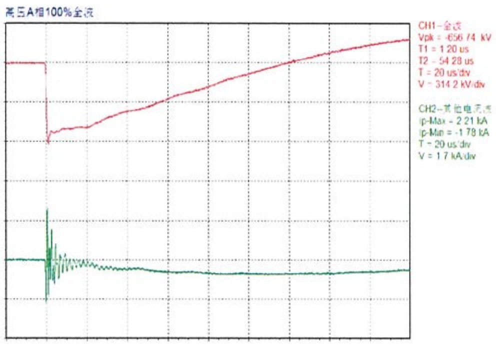




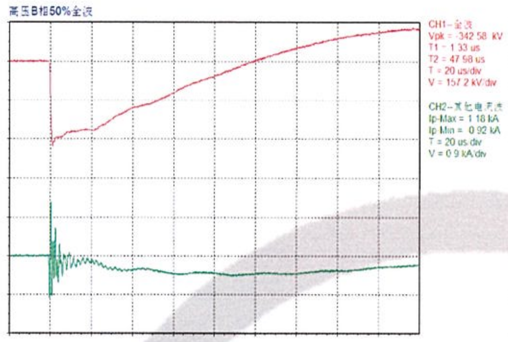


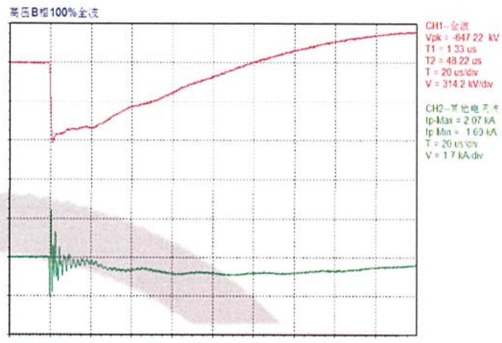


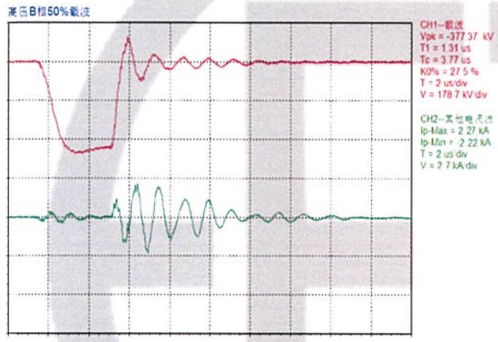


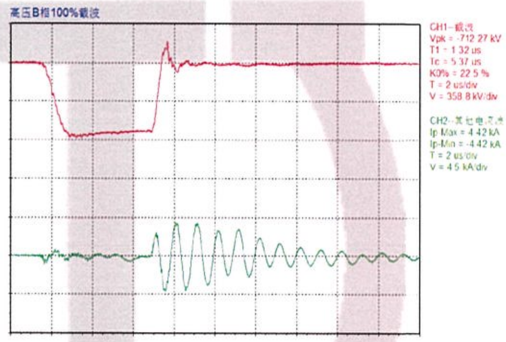


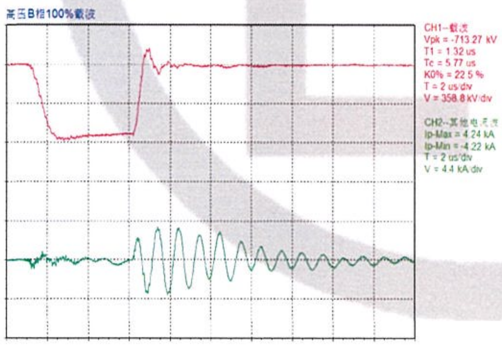
Tested terminal: B Test polarity: negative Channel 1: voltage wave Channel 2: current wave

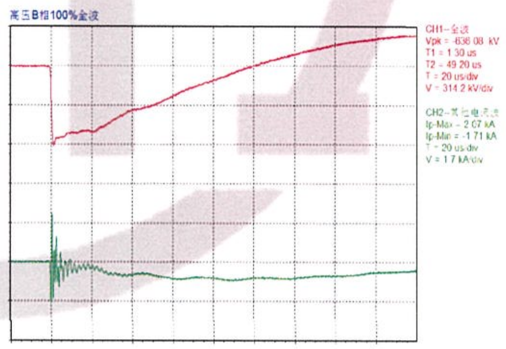


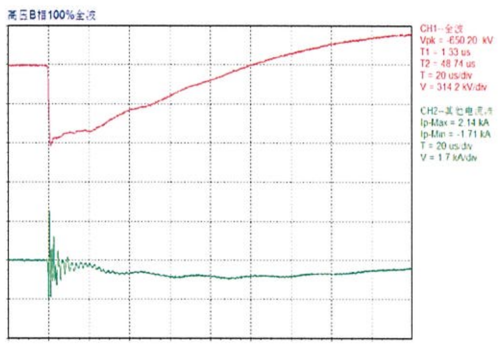




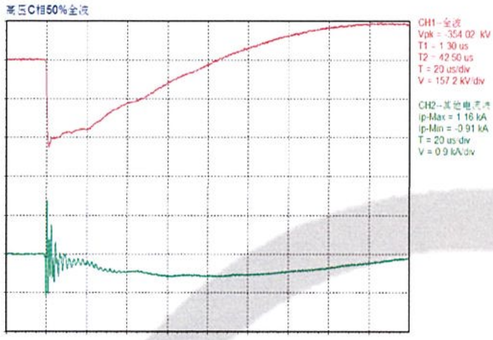


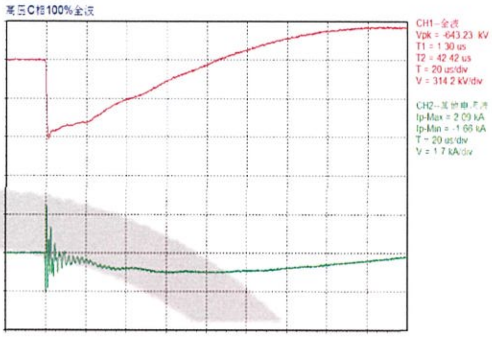


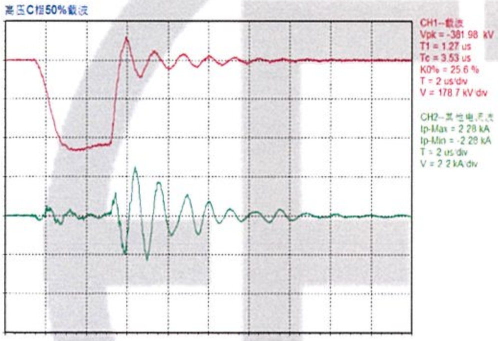


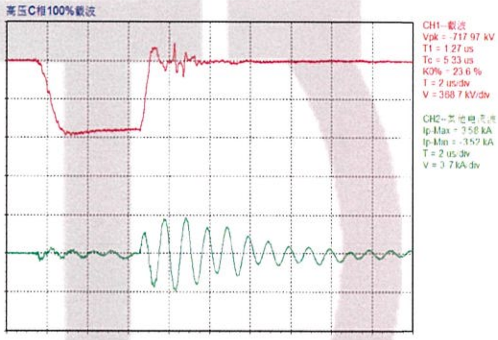


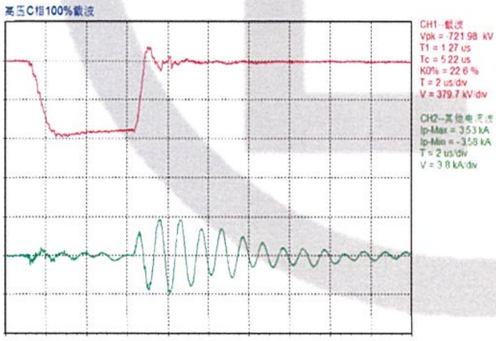
Tested terminal: C Test polarity: negative Channel 1: voltage wave Channel 2: current wave

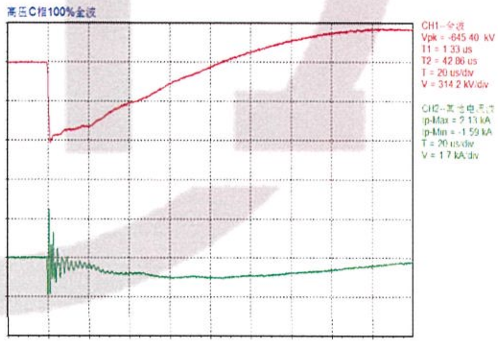


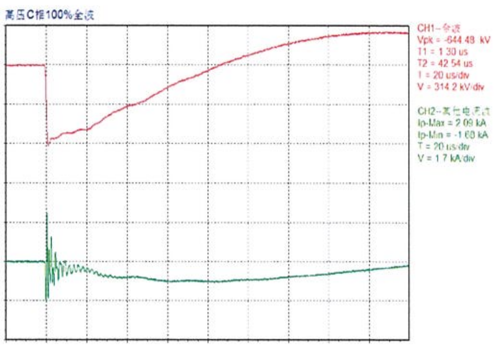




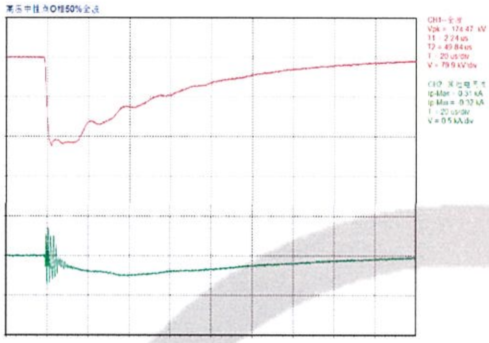


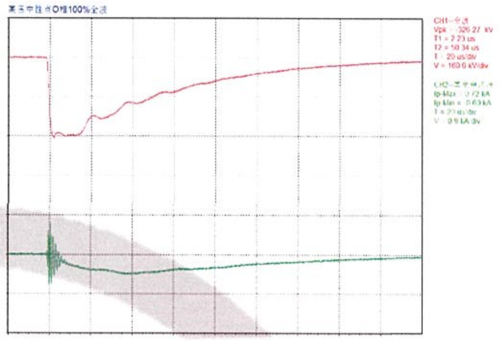


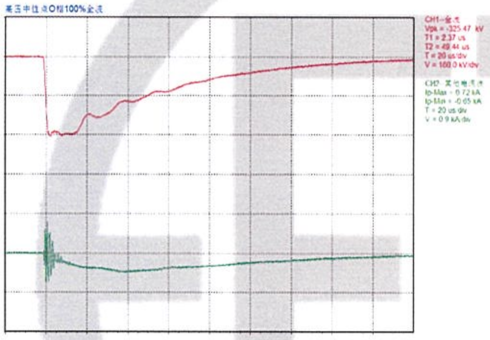


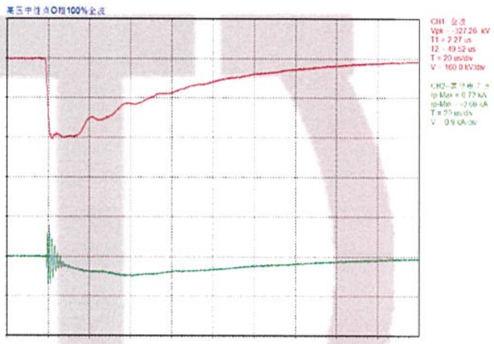


Tested terminal: O Test polarity: negative Channel 1: voltage wave Channel 2: current wave









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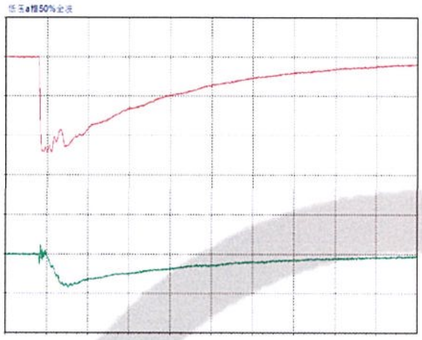
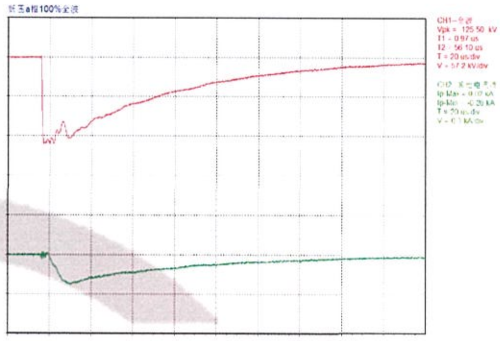
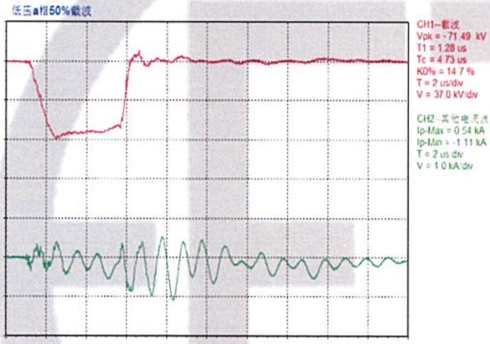
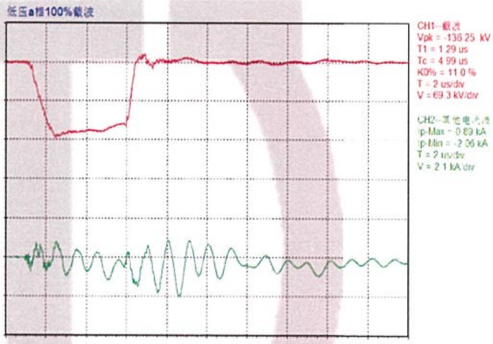
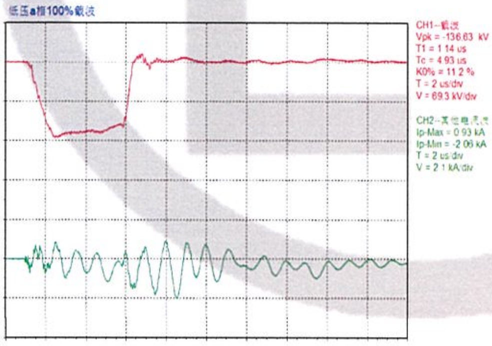
Tested terminal: a Test polarity: negative Channel 1: voltage wave Channel 2: current wave

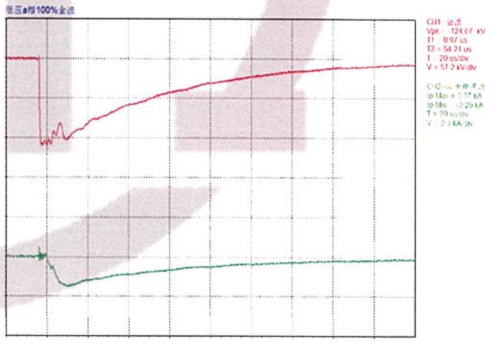


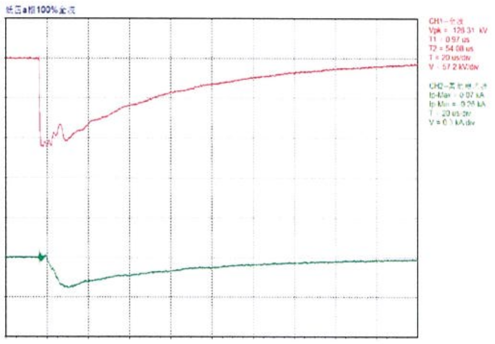




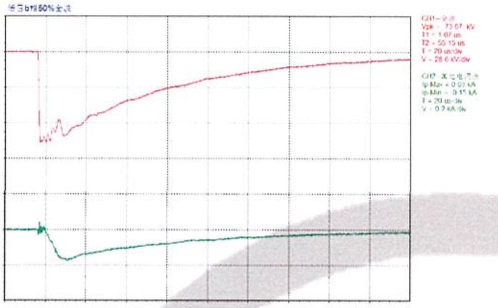


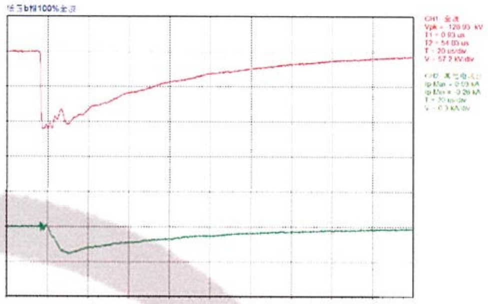


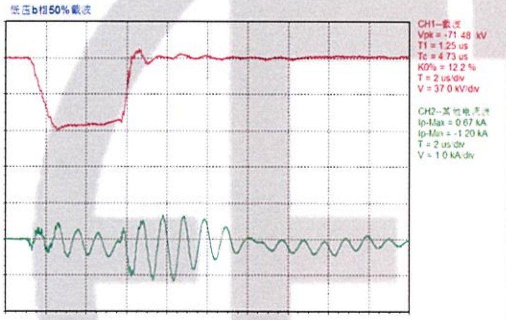


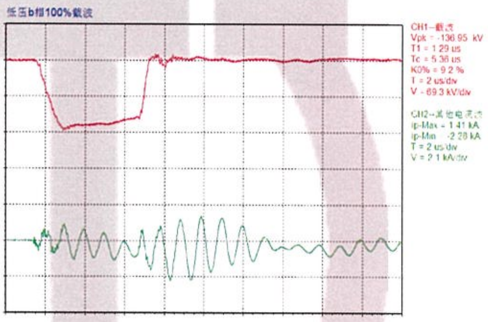


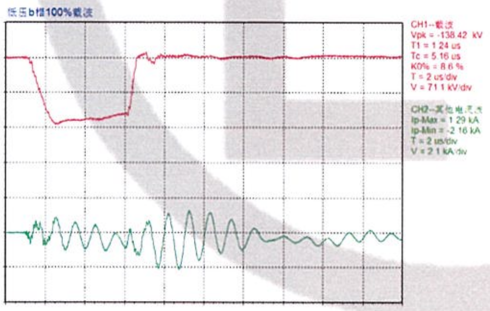
Tested terminal: b Test polarity: negative Channel 1: voltage wave Channel 2: current wave

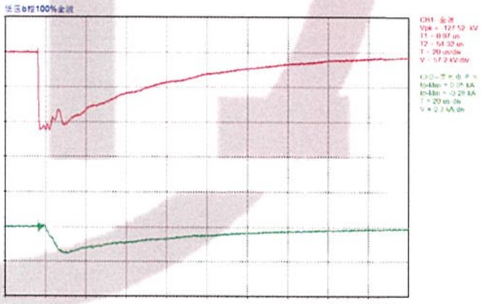


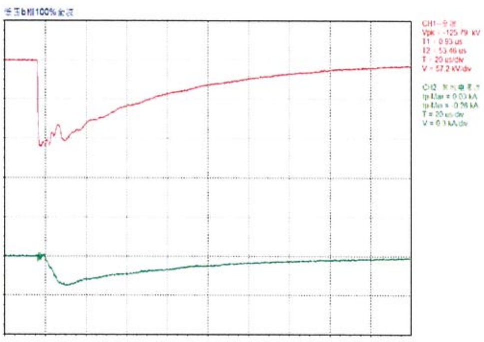




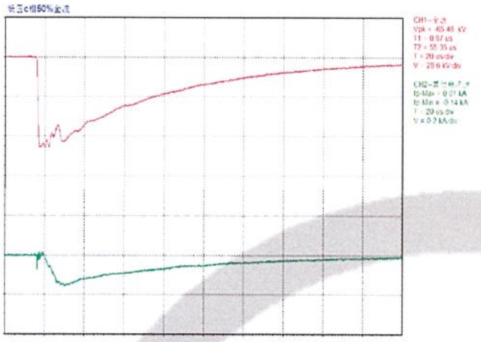


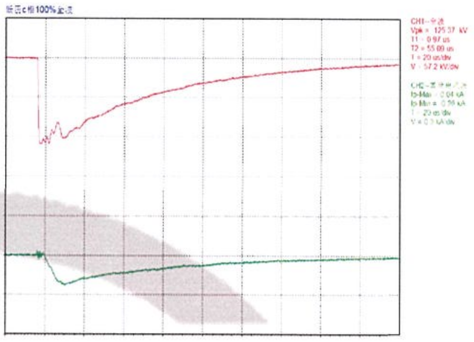


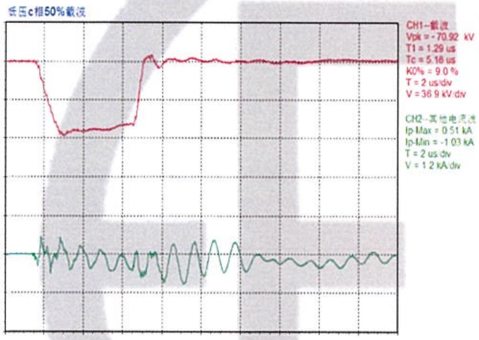


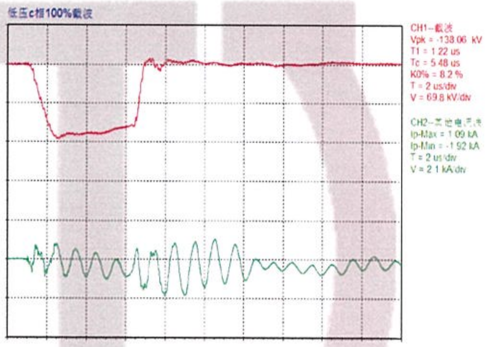


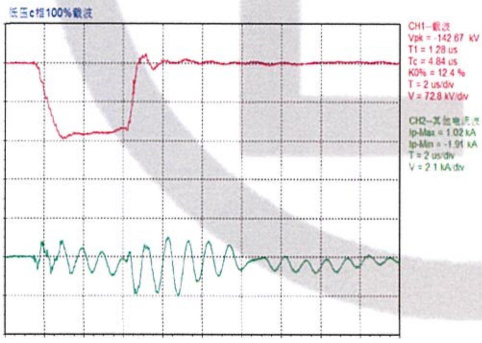
Tested terminal: c   Test polarity: negative   Channel 1: voltage wave   Channel 2: current wave

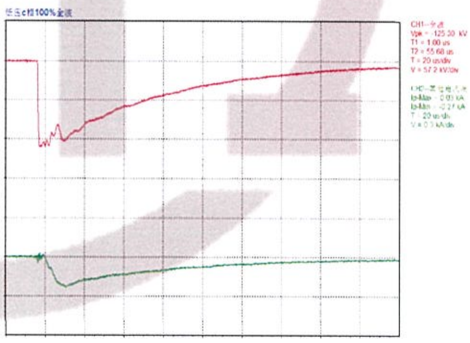


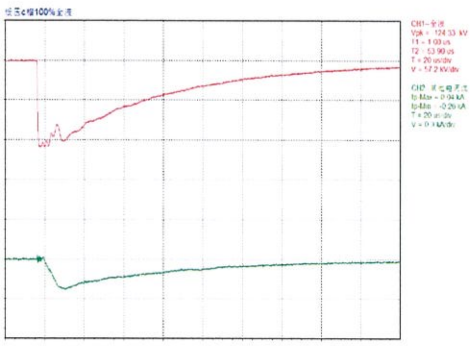












4.14 Tests on on-load tap-changers (routine test)

Test date: Aug. 15, 2021

Operation tests on on-load tap-changers:

a. with the transformer de-energized, eight complete cycles of operation (a cycle of operation goes from one end of the tapping range to the other, and back again);

b. with the transformer de-energized, and with the auxiliary voltage reduced to 85% of its rated value, one complete cycle of operation;

c. with the transformer energized at rated voltage and frequency at no load, one complete cycle of operation;

d. with one winding short-circuited and, as far as practicable, rated current in the tapped winding, 10 cycles of tap-change operations across the range of two steps on each side from where a coarse or reversing changeover selector operates, or otherwise from the middle tapping (the tapchanger will pass 20 times through the changeover position).

Test result: PASS.

4.15 Applied voltage test (AV) (routine test)

Test date: Aug. 17, 2021

Relative humidity: 58%; Ambient temperature: 32.0°C; Oil temperature: 33.0°C; Air pressure: 102kPa

Applied voltage parts	Test voltage (kV)	Test duration (s)	Result
HV neutral point—LV and earth	140.0	60	PASS
LV—HV and earth	55.0	60	

4.16 Line terminal AC withstand test (LTAC) (routine test)

Test date: Aug. 17, 2021

Relative humidity: 58%; Ambient temperature: 32.0°C; Oil temperature: 33.0°C; Air pressure: 102kPa

Phase to earth test

Applied voltage parts	Tap position	Applied voltage (kV)	Induced voltage (kV)		Frequency (Hz)	Duration (s)	Results
		LV	HV				
a-c	1	43.92	A	275	200	30	PASS
a-b		43.92	B	275			
b-c		43.92	C	275			

4.17 Induced voltage withstand test and induced voltage test with partial discharge measurement (IVW, IVPD) (routine test)

Test date: Aug. 17, 2021

Relative humidity: 58%; Ambient temperature: 32.0°C; Oil temperature: 33.0°C; Air pressure: 102kPa

HV tapping position 10, frequency 200Hz.

Applied voltage		Duration (min)	Partial discharge magnitude (pC)		
Multiple	Phase-earth voltage (kV)		A	B	C
$0.4Ur/\sqrt{3}$	30.5	/	<20	<20	<20
$1.2Ur/\sqrt{3}$	91.5	1	<40	<40	<40
$1.58Ur/\sqrt{3}$	120.4	5	<60	<60	<60
$2.0Ur/\sqrt{3}$	152.4	0.5	/	/	/
$1.58Ur/\sqrt{3}$	120.4	5	<60	<60	<60
		10	<60	<60	<60
		15	<60	<60	<60
		20	<60	<60	<60
		25	<60	<60	<60
		30	<60	<60	<60
		35	<60	<60	<60
		40	<50	<50	<50
		45	<50	<50	<50
		50	<50	<50	<50
		55	<50	<50	<50
		60	<50	<50	<50
$1.2Ur/\sqrt{3}$	91.5	1	<50	<50	<50
$0.4Ur/\sqrt{3}$	30.5	/	<20	<20	<20

Remark: Ur=132kV.

4.18 Insulating liquid test, measurement of dissolved gasses in dielectric liquid from each separate oil compartment except diverter switch compartment (routine test, type test, commission test)

Test date: Aug. 16, 2021
Relative humidity: 55%; Ambient temperature: 32.5°C

Dielectric dissipation factor (90°C)	Breakdown voltage (kV)	Water content (mg/L)	Flash point (closed cup) (°C)
0.11%	61.8	7.3	175

Gas chromatograph analysis (before all tests)

Test date: Aug. 16, 2021
μL/L

H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
0.89	3.87	65.38	0.23	0	0	0	0.23

Gas chromatograph analysis (after dielectric test)

Test date: Aug. 17, 2021
μL/L

H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
0.94	4.83	73.87	0.34	0	0	0	0.34

Gas chromatograph analysis (after long-duration no-load test)

Test date: Aug. 18, 2021
μL/L

H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
1.23	3.98	76.38	0.41	0	0	0	0.41

Gas chromatograph analysis (after temperature-rise test)

Test date: Aug. 19, 2021
μL/L

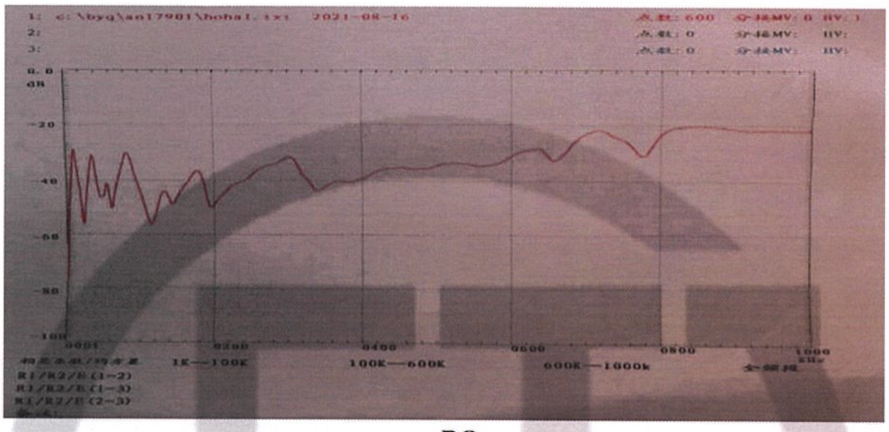
H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
1.54	5.89	89.63	0.52	0	0	0	0.52

4.19 Measurement of frequency response (special test)

Test date: Aug. 16, 2021

HV winding frequency response curves (Tap 1)

AO



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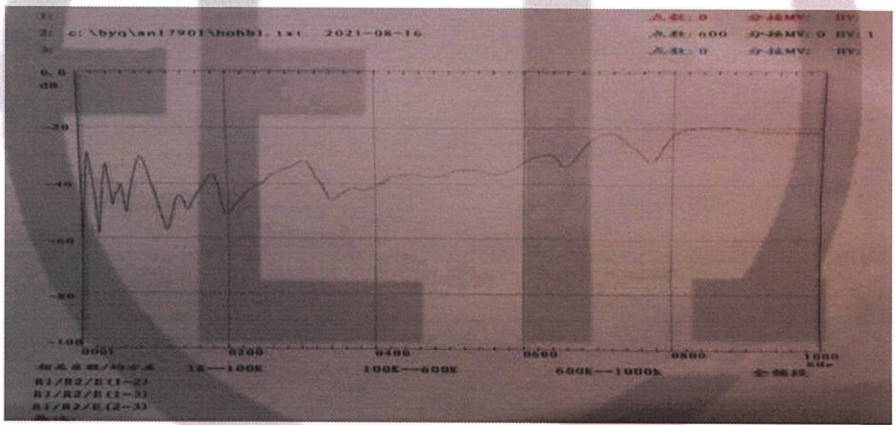

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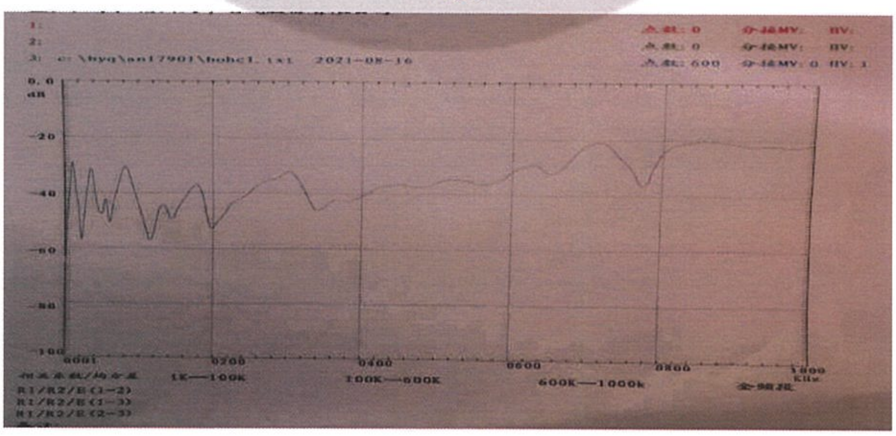
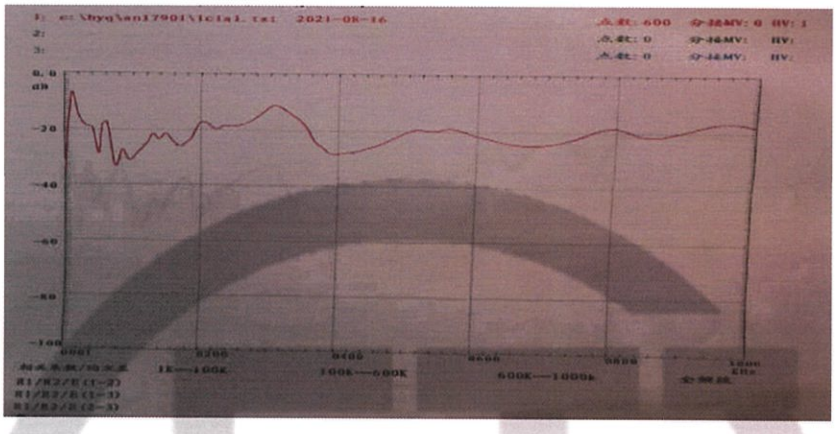


LV winding frequency response curves (Tap 1)

ab



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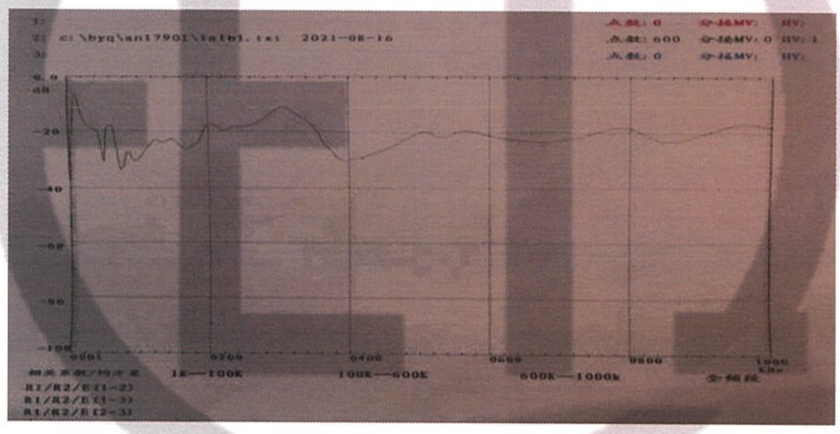

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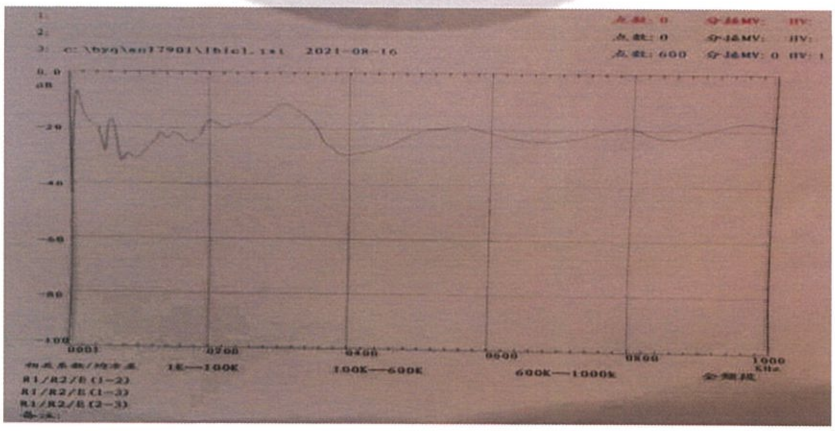
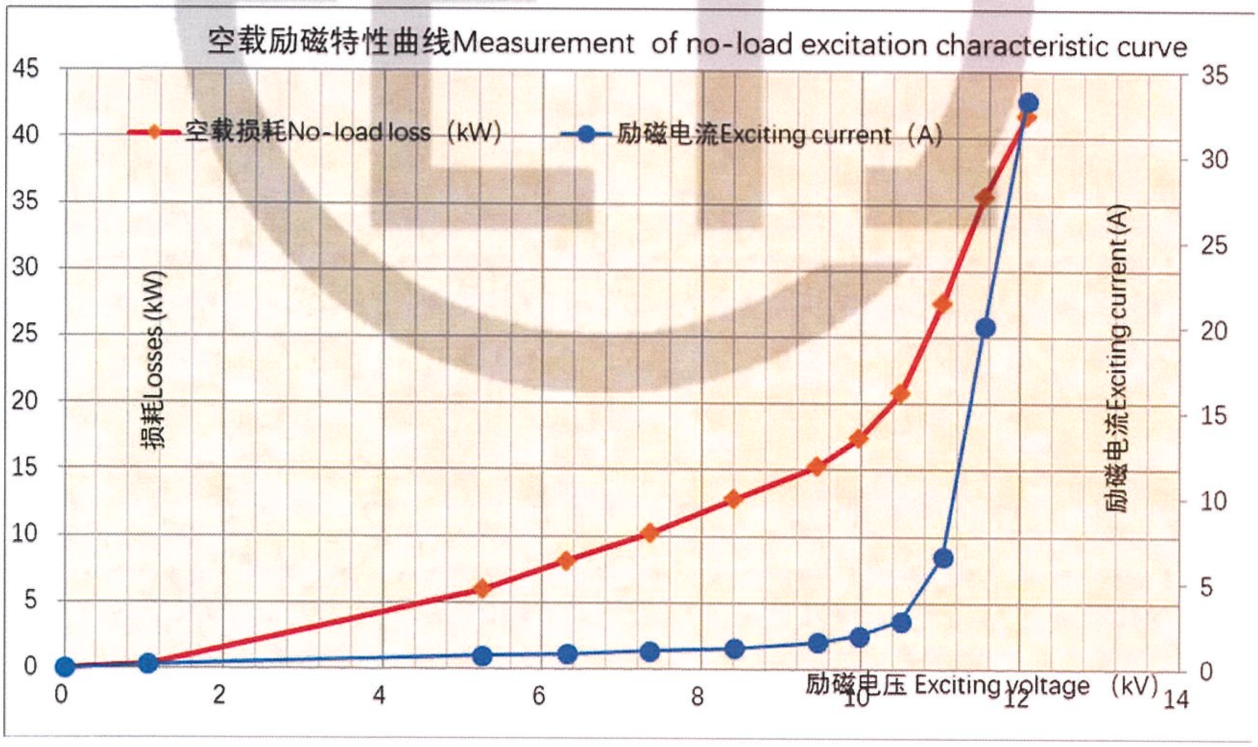


4.20 Measurement of no-load excitation characteristics (commission test)

Test date: Aug. 16, 2021

Voltage	Measured voltage (kV)	Measured current (A)	Measured loss (kW)
$0.10U_r$	0.213	0.286	0.857
$0.50U_r$	10.510	0.811	12.787
$0.60U_r$	12.598	0.946	18.119
$0.70U_r$	14.703	1.097	24.509
$0.80U_r$	16.806	1.290	31.991
$0.90U_r$	18.905	1.612	42.035
$0.95U_r$	19.954	1.923	48.487
$1.00U_r$	21.008	2.790	58.499
$1.05U_r$	22.057	6.914	74.915
$1.10U_r$	23.102	30.93	99.126
$1.15U_r$	24.148	40.84	101.120

Remarks: U_r=21kV.



4.21 Long-duration no-load test (commission test)

Test date: from Aug. 17, 2021 to Aug. 18, 2021

Test with 1.1 times rated voltage applied on LV side for 12h. No C_2H_2 is found in oil and the content of total hydrocarbon has no obvious variation before and after the test. For gas chromatograph analysis data, see 4.18.

Duration (h)	Voltage (kV)	Current (A)	No-load loss (kW)
1	23.103	30.92	99.137
2	23.112	30.96	99.154
3	23.115	31.05	99.134
4	23.107	30.98	99.146
5	23.104	30.95	99.131
6	23.104	30.94	99.128
7	23.107	30.98	99.123
8	23.103	30.93	99.120
9	23.097	30.98	99.118
10	23.098	30.90	99.116
11	23.095	30.87	99.117
12	23.104	30.86	99.111

4.22 Measurement of zero-sequence impedances on three-phase transformers (special test)

Test date: Aug. 16, 2021

Connection symbol	Power supply terminal	Open-circuit terminal	Short-circuit terminal	Tapping position	Applied current (A)	Measured voltage (V)	Impedance (Ω)
YNd11	ABC-O	a, b, c	/	10	233.08	1254.1	16.14

4.23 Measurement of the harmonics of the no-load current (commission test)

Test date: Aug. 16, 2021

No	CH-A THD =43.82		CH-B THD =38.74		CH-C THD =52.08	
	In(A)	In/I1(%)	In(A)	In/I1(%)	In(A)	In/I1(%)
01	2.255	100.00	3.022	100.00	2.099	100.00
02	0.020	0.902	0.021	0.703	0.011	0.517
03	0.500	22.180	0.196	6.484	0.616	29.369
04	0.015	0.671	0.016	0.523	0.010	0.482
05	0.704	31.234	0.974	32.230	0.754	35.929
06	0.005	0.198	0.005	0.151	0.002	0.080
07	0.427	18.919	0.572	18.928	0.447	21.315
08	0.009	0.409	0.007	0.227	0.006	0.295
09	0.098	4.338	0.004	0.131	0.099	4.739
10	0.005	0.242	0.006	0.184	0.002	0.114
11	0.169	7.502	0.201	6.650	0.158	7.536
12	0.001	0.063	0.001	0.040	0.002	0.076
13	0.087	3.872	0.115	3.812	0.096	4.569
14	0.002	0.078	0.002	0.051	0.001	0.022
15	0.017	0.761	0.002	0.061	0.017	0.813
16	0.003	0.111	0.002	0.052	0.001	0.034
17	0.035	1.557	0.414	1.370	0.031	1.511
18	0.002	0.090	0.001	0.046	0.001	0.068
19	0.017	0.747	0.023	0.754	0.020	0.957
20	0.002	0.099	0.002	0.064	0.002	0.072

4.24 Temperature-rise test (including calculation of the winding hot-spot temperature-rise) (type test)

4.24.1 In an ONAN condition Test date: Aug. 18, 2021

The method of temperature rise is the equivalent test in short-circuit connection, the HV (tapping position 19) is supplying power, the LV is short-circuited, test duration is 11h and of which the stabilization time is 3h.
Measurement of top oil temperature-rise:
During the test, it is required to apply 404.252kW of total loss and 404.650kW of total loss is actually applied.
Measurement of winding temperature-rise:
During the test, 617.73A current is required and 617.48A current is actually applied.

Measured data

<table>
<tr><th colspan="3">Top oil temperature-rise and average oil temperature-rise</th><th colspan="7">Measurement of average temperature-rise windings to oil</th><th colspan="2">Ambient temperature (°C)</th></tr>
<tr><th>Top oil temperature (°C)</th><th>Average oil temperature (°C)</th><th>Total loss injection/ specified total loss (%)</th><th>Current injection/ rated current (%)</th><th colspan="2">Cold resistance (Ω)</th><th colspan="2">Average oil temperature (°C)</th><th colspan="2">Average winding temperature (°C)</th><th>Total loss</th><th>Measurement of cold resistance</th></tr>
<tr><td rowspan="2">80.9</td><td rowspan="2">67.2</td><td rowspan="2">100.10</td><td rowspan="2">99.96</td><td>HV</td><td>0.1489</td><td>At the instant of shutdown</td><td>64.2</td><td>HV</td><td>85.3</td><td rowspan="2">31.4</td><td rowspan="2">33.0</td></tr>
<tr><td>LV</td><td>6.537×10^{-3}</td><td>At the end of cooling</td><td>57.4</td><td>LV</td><td>85.3</td></tr>
</table>

Calculations of temperature-rise

<table>
<tr><td colspan="2">Top oil temperature-rise (K)</td><td>49.7</td></tr>
<tr><td rowspan="2">Winding temperature-rise (K)</td><td>HV</td><td>57.0</td></tr>
<tr><td>LV</td><td>57.0</td></tr>
<tr><td rowspan="2">Winding hot-spot temperature-rise (K)</td><td>HV</td><td>72.0</td></tr>
<tr><td>LV</td><td>71.8</td></tr>
<tr><td colspan="2">Temperature-rise of tank surface and mental structural parts (K)</td><td>54.7</td></tr>
</table>

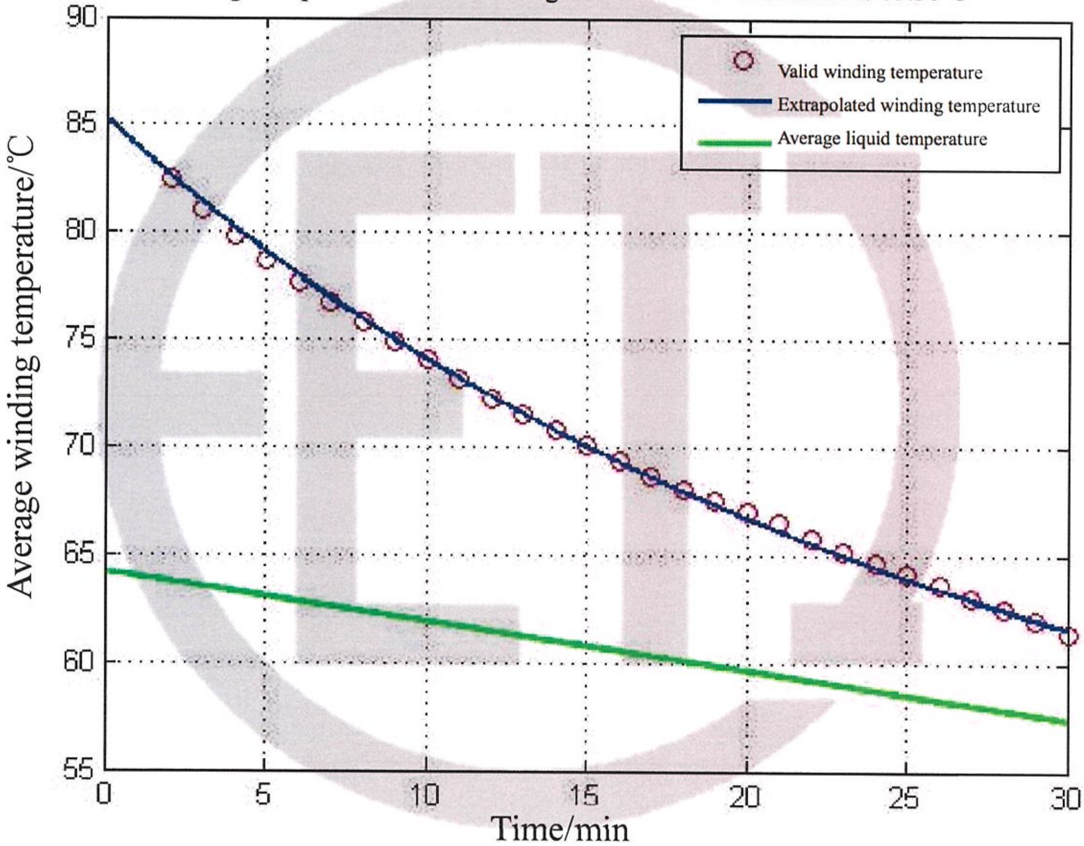
Remarks: the calculated of temperature-rise are the corrected value under specified total loss and rated current.
HV winding hot-spot coefficient is 1.06, LV winding hot-spot coefficient is 1.05.

Winding temperature curve

Average winding temperature data

Average HV winding temperature	85.3°C

Average temperature of HV winding at the instant of shut down is 85.30°C

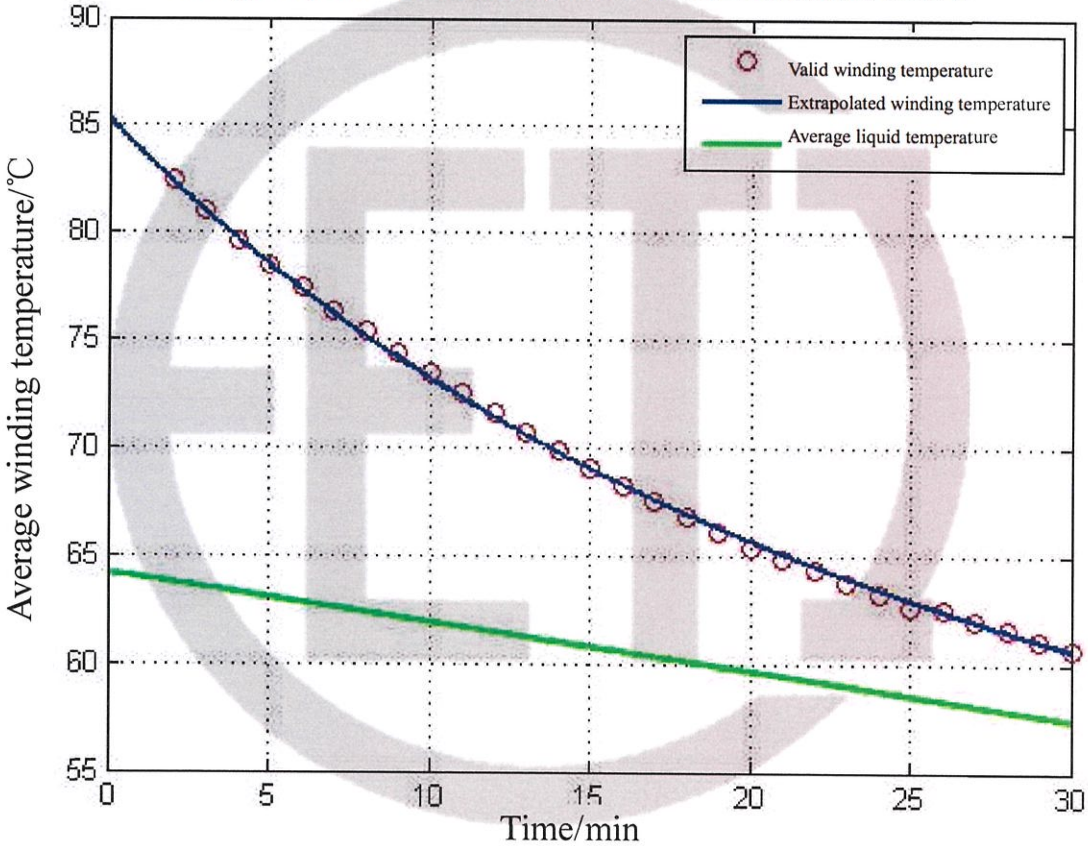


Winding temperature curve

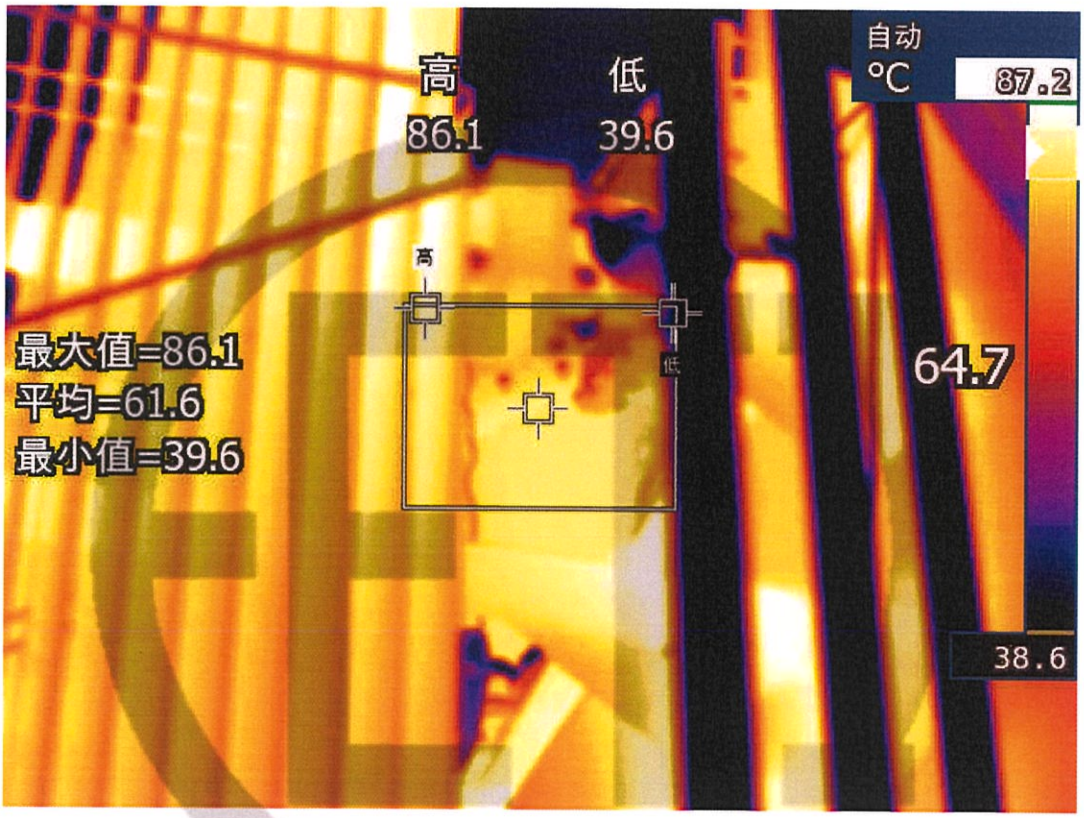
Average winding temperature data

Average LV winding temperature	85.3℃

Average temperature of LV winding at the instant of shut down is 85.30℃



Thermograph of tank surface and structural parts temperature-rise



4.24.2 In an ONAF condition

Test date: from Aug. 18, 2021 to Aug. 19, 2021

The method of temperature rise is the equivalent test in short-circuit connection, the HV (tapping position 19) is supplying power, the LV is short-circuited, test duration is 8h and of which the stabilization time is 3h.

Measurement of top oil temperature-rise:

During the test, it is required to apply 598.740kW of total loss and 598.390kW of total loss is actually applied.

Measurement of winding temperature-rise:

During the test, 772.16A current is required and 772.24A current is actually applied.

Measured data

Top oil temperature-rise and average oil temperature-rise			Measurement of average temperature-rise windings to oil							Ambient temperature (°C)	
Top oil temperature (°C)	Average oil temperature (°C)	Total loss injection/ specified total loss (%)	Current injection/ rated current (%)	Cold resistance (Ω)		Average oil temperature (°C)		Average winding temperature (°C)		Total loss	Measurement of cold resistance
82.1	66.9	99.94	100.01	HV	0.1489	At the instant of shutdown	63.5	HV	83.6	31.5	33.0
				LV	6.537×10^{-3}	At the end of cooling	55.8	LV	84.0		

Calculations of temperature-rise

Top oil temperature-rise (K)		50.2
Winding temperature-rise (K)	HV	55.4
	LV	55.8
Winding hot-spot temperature-rise (K)	HV	71.8
	LV	72.0
Temperature-rise of tank surface and mental structural parts (K)		55.3

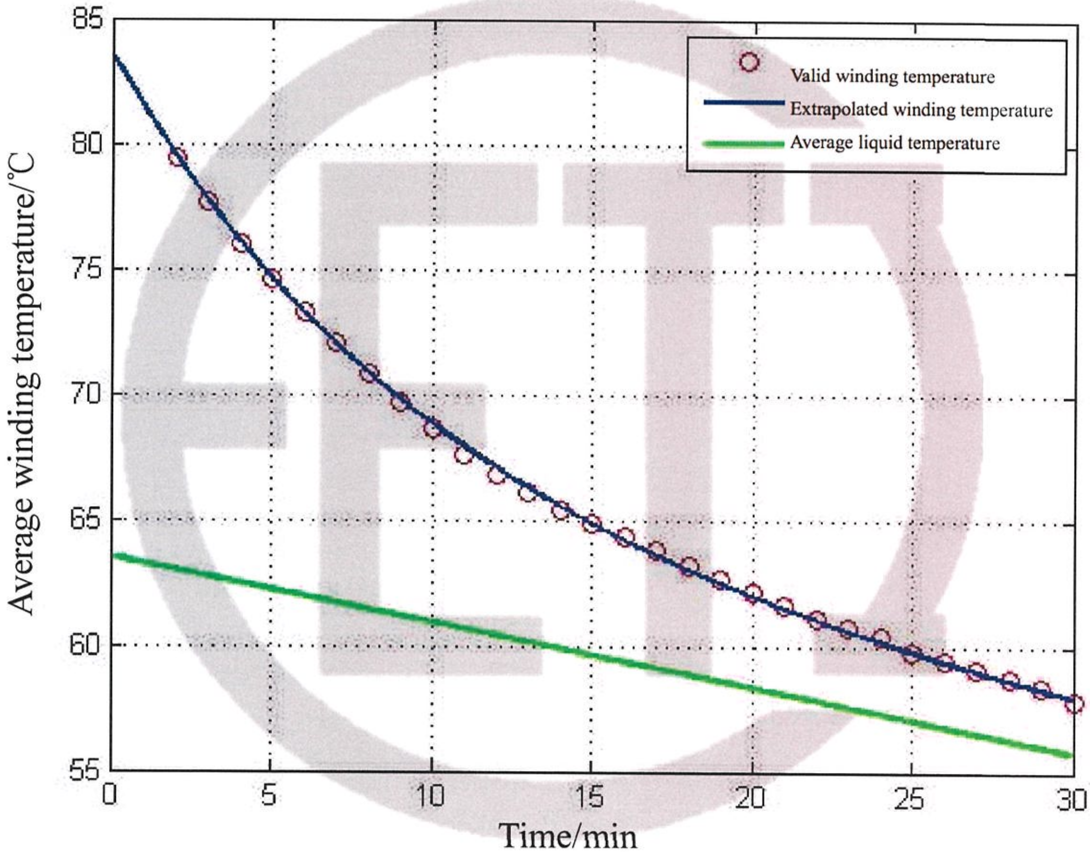
Remarks: the calculated of temperature-rise are the corrected value under specified total loss and rated current. HV winding hot-spot coefficient is 1.06, LV winding hot-spot coefficient is 1.05.

Winding temperature curve

Average winding temperature data

Average HV winding temperature	83.6°C

Average temperature of HV winding at the instant of shut down is 83.63°C

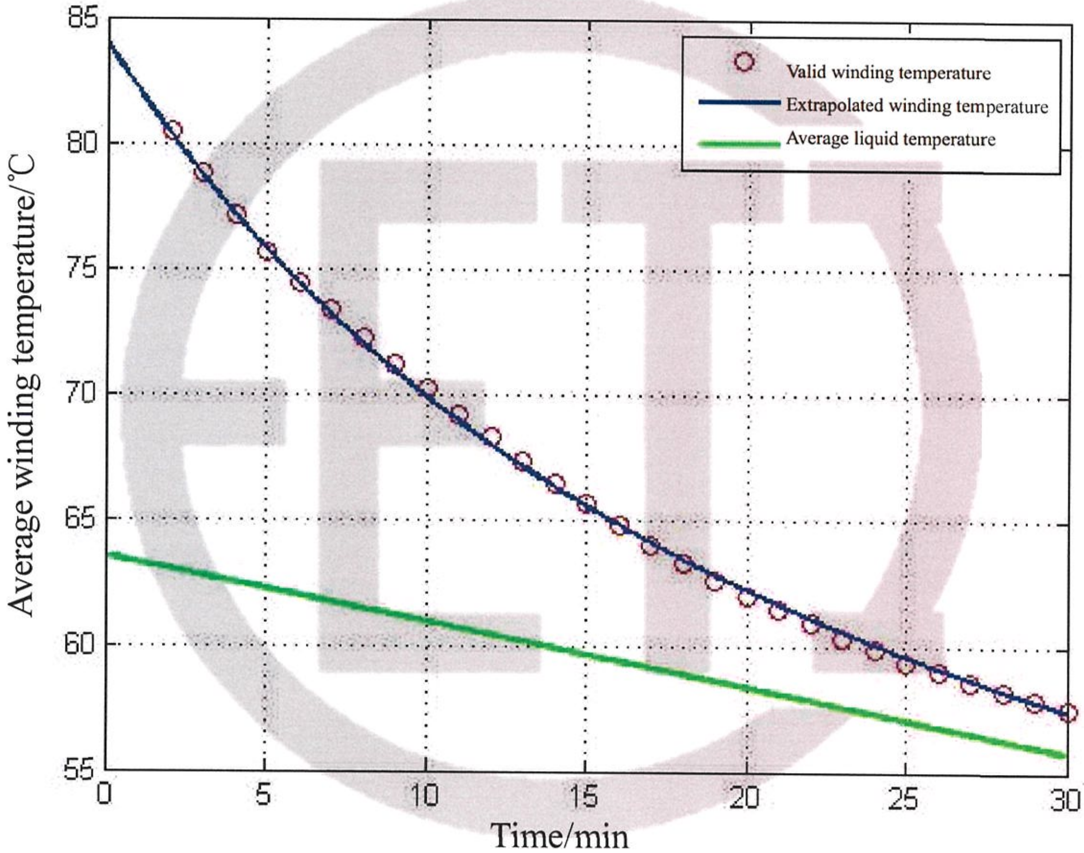


Winding temperature curve

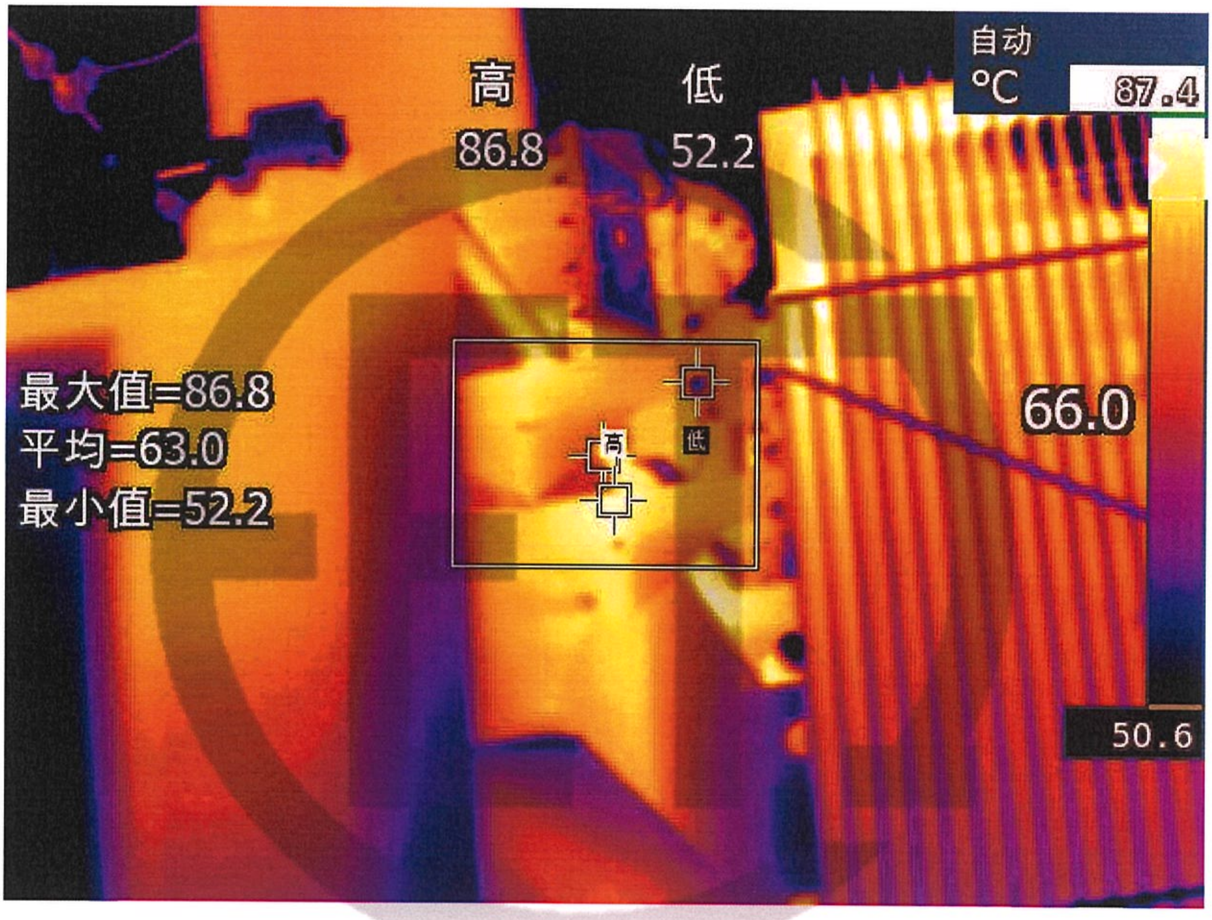
Average winding temperature data

Average LV winding temperature	84.0°C

Average temperature of LV winding at the instant of shut down is 83.99°C



Thermograph of tank surface and structural parts temperature-rise



4.25 Leak testing with pressure for liquid-immersed transformers (routine test)

Test date: from Aug. 13, 2021 to Aug. 15, 2021

Test method	Parts of applied voltage	Applied pressure (kPa)	Duration (h)	Result
Static pressure method	Body	30.0	24	No oil leakage or damage
	On-load tap-changer tank	30.0	24	No oil leakage or damage

Remarks: the product is a general tank, there is no oil leakage or damage before leak testing.

4.26 Determination of sound levels (type test)

Test date: Aug. 17, 2021

4.26.1 In an ONAN condition

4.26.1.1 Rough estimation of the load current sound power level

Equation:

$$L_{WA,IN} \approx 39 + 18\lg\frac{Sr}{Sp} = 78.2\text{dB(A)}$$

where: Sr—the rated power is 150MVA;

Sp—the reference power is 1MVA.

Because $L_{WA,IN}$ is less 11.8dB(A) than limit value 90dB(A) of assured sound power level, according to standard requirement, it doesn't need to measure load current sound power level.

4.26.1.2 Measurement of sound pressure level and calculation of sound power level at no-load condition

The transformer is rated excitation, the prescribed contour shall be spaced 0.3m away from the principal radiating surface, the distance between measured points is 0.94m, the number of measured points is 30, the height 1 of measured point is 1.5m, the height 2 of measured point is 3.0m.

Test environment

The total area of the surface of the test room Sv (m^2)	The average acoustic absorption coefficient α	The amount of acoustic absorption A (m^2)	Distance from the principal radiating surface (m)	The area of the measurement surface S (m^2)	Environmental correction K (dB)
5710	0.15	856.5	0.3	159.3	2.4

The results of measurement (dB)

Status of cooling device	Average of background noise		Average noise value of transformer $\overline{L_{PA0}}$	A weighted sound pressure level $\overline{L_{pA}} = 10\lg\left(10^{0.1\overline{L_{pA0}}} - 10^{0.1\overline{L_{bgA}}}\right) - K$	A weighted sound power level $L_{WA} = \overline{L_{pA}} + 10\lg(S/S_0)$
	Before the test	After the test			
/	43.9	44.2	60.7	58	80

Remarks: $\overline{L_{PA0}}$: the uncorrected average A-weighted sound pressure level; $\overline{L_{PA0}} = 10\lg\left(\frac{1}{N}\sum_{i=1}^{N}10^{0.1L_{PAi}}\right)$

$\overline{L_{bgA}}$: the lower of the two calculated average A weighted background noise pressure level.

4.26.2 In an ONAF condition Test date: Aug. 17, 2021

4.26.2.1 Rough estimation of the load current sound power level

Equation:

$$L_{WA,IN} \approx 39 + 18\lg\frac{Sr}{Sp} = 78.2\text{dB(A)}$$

where: Sr—the rated power is 150MVA;
Sp—the reference power is 1MVA.

Because $L_{WA,IN}$ is less 11.8dB(A) than limit value 90dB(A) of assured sound power level, according to standard requirement, it doesn't need to measure load current sound power level.

4.26.2.2 Measurement of sound pressure level and calculation of sound power level at no-load condition

The transformer is rated excitation, the prescribed contour shall be spaced 2.0m away from the principal radiating surface, the distance between measured points is 0.99m, the number of measured points is 40, the height 1 of measured point is 1.5m, the height 2 of measured point is 3.0m.

Test environment

The total area of the surface of the test room Sv (m^2)	The average acoustic absorption coefficient α	The amount of acoustic absorption A (m^2)	Distance from the principal radiating surface (m)	The area of the measurement surface S (m^2)	Environmental correction K (dB)
5710	0.15	856.5	2.0	214.5	3.0

The results of measurement (dB)

Status of cooling device	Average of background noise		Average noise value of transformer $\overline{L_{PA0}}$	A weighted sound pressure level $\overline{L_{pA}} = 10\lg\left(10^{0.1\overline{L_{PA0}}} - 10^{0.1\overline{L_{bgA}}}\right) - K$	A weighted sound power level $L_{WA} = \overline{L_{pA}} + 10\lg(S/S_0)$
	Before the test	After the test			
/	43.2	43.7	61.8	59	82

Remarks: $\overline{L_{PA0}}$: the uncorrected average A-weighted sound pressure level; $\overline{L_{PA0}} = 10\lg\left(\frac{1}{N}\sum_{i=1}^{N}10^{0.1L_{PAi}}\right)$

$\overline{L_{bgA}}$: the lower of the two calculated average A weighted background noise pressure level.

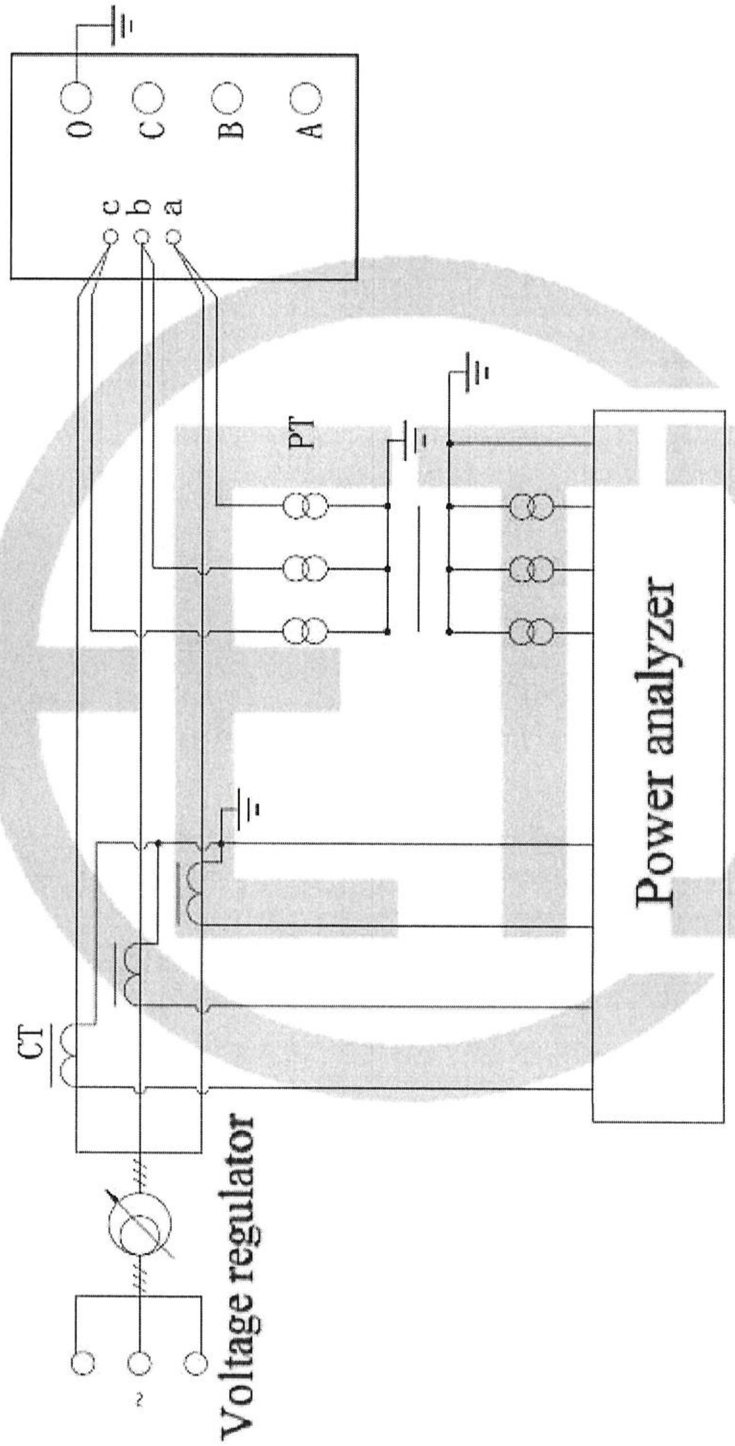
4.27 Mechanical strength test of tank (type test)

Test date: Aug. 11, 2021

Test method	Applied pressure (kPa)	Duration (min)
Vacuum degree	0.133	10
Positive pressure	100	10

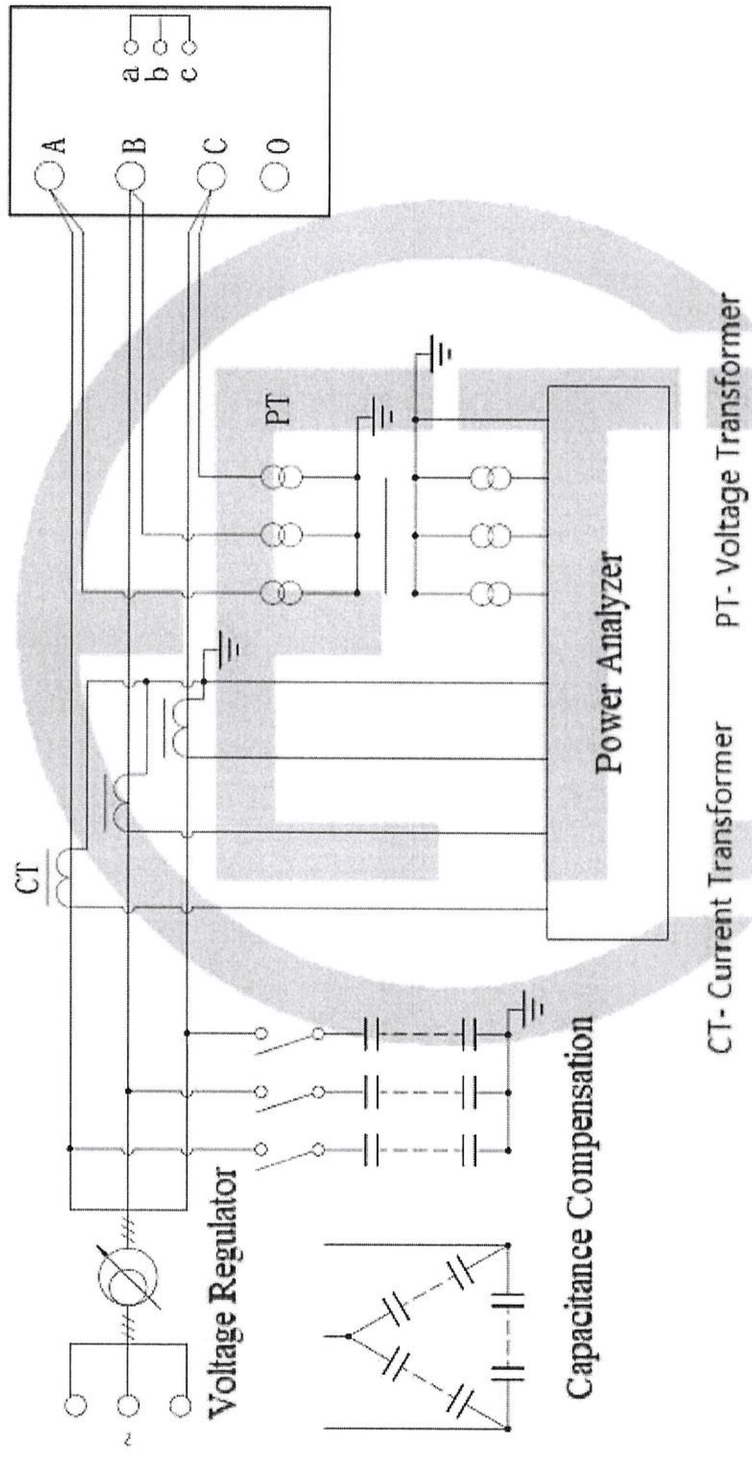
Measured items		Measured points												
		Tank wall										Tank cover		
		HV side			LV side			Left side		Right side		Middle of length		
Vacuum degree	Initial distance (mm)	223	223	228	234	228	225	197	294	214	219	241	244	245
	Distance after pressure injection (mm)	230	229	237	240	238	232	206	301	224	227	247	253	252
	Distance without pressure (mm)	224	224	229	235	229	225	198	295	216	221	242	246	246
	Elastic deformation (mm)	7	6	9	6	10	7	9	7	10	8	6	9	7
	Permanent deformation (mm)	1	1	1	1	2	0	1	1	2	2	1	2	1
Positive pressure	Initial distance (mm)	224	224	229	235	229	225	298	295	216	221	242	246	246
	Distance after pressure injection (mm)	216	215	224	229	223	217	291	286	206	214	233	239	238
	Distance without pressure (mm)	224	222	228	234	227	223	297	294	215	221	241	245	244
	Elastic deformation (mm)	8	11	5	6	6	8	7	9	10	7	9	7	8
	Permanent deformation (mm)	0	2	1	1	2	2	1	1	1	0	1	1	2
Result		No damage												

Remarks: 1. the product is a general tank.
2. the described left and right sides of test point are viewed from HV side.
3. the left, middle and right test points of HV and LV side are obtained from the 1/2 height in vertical direction, 1/4, 1/2 and 3/4 position respectively in horizontal direction.

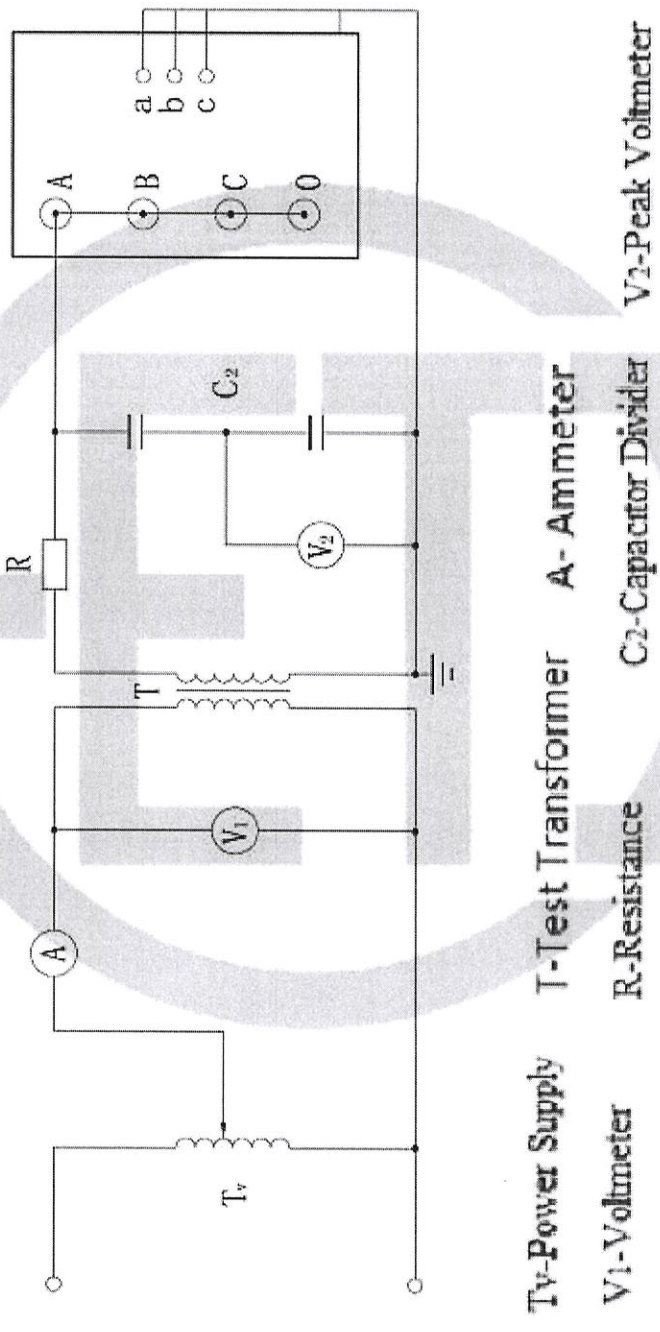


CT- Current transformer PT-Voltage transformer

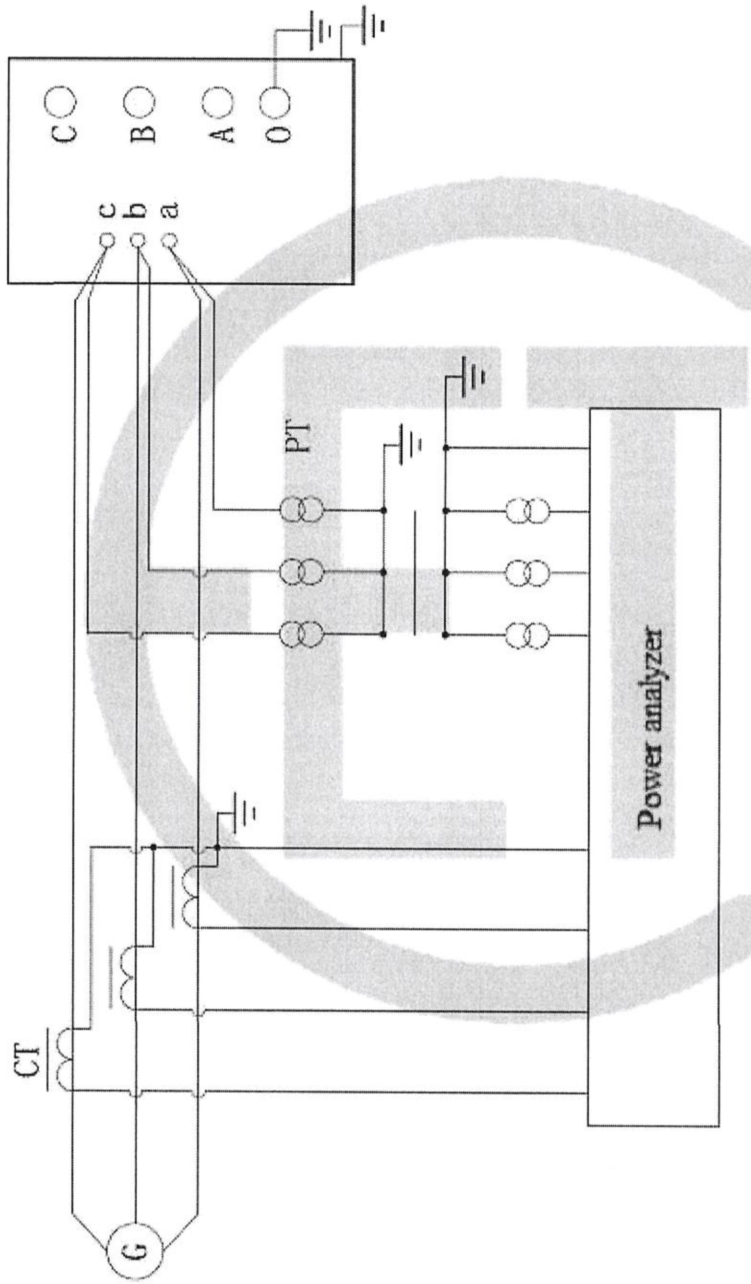
Schematic diagram of measurement of no-load loss and current



Schematic diagram of measurement of short-circuit impedance and load loss

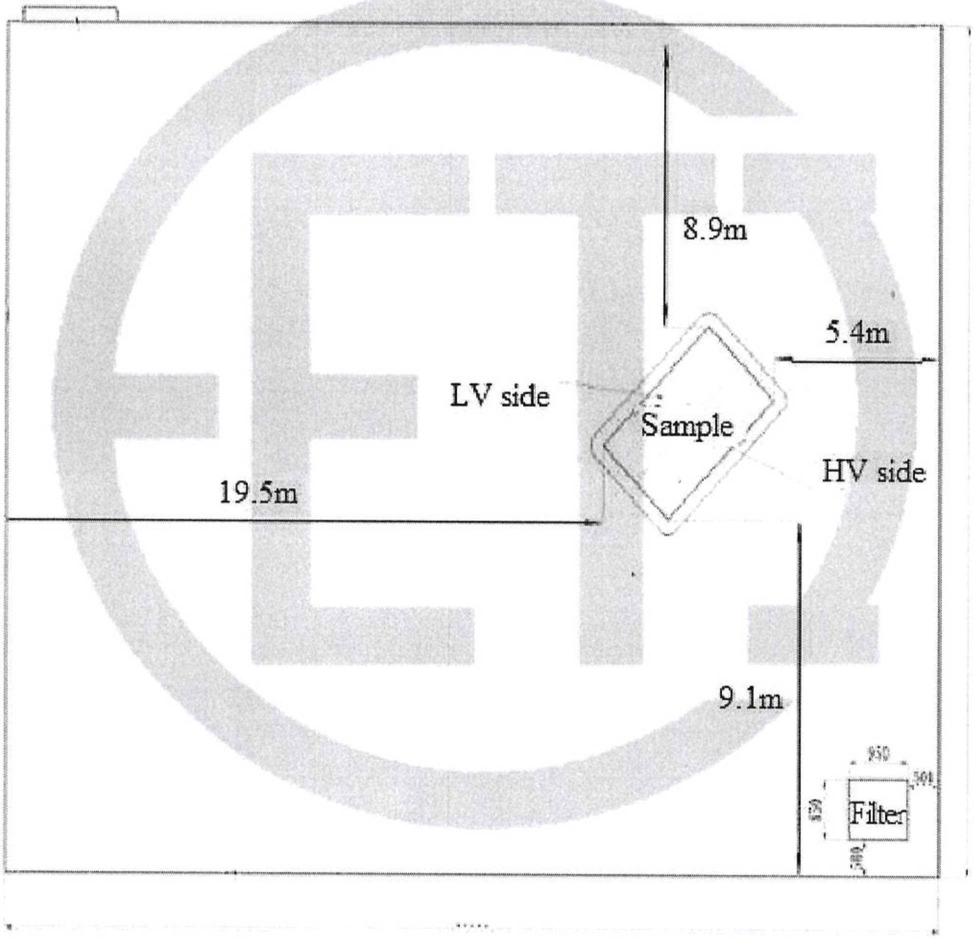


Schematic diagram of applied voltage test



G-Induction power supply CT-Current transformer PT-Voltage transformer

Schematic diagram of induced voltage withstand test



Sound levels arrangement diagram

Instruments used in the tests

No	Test items	Name & type of instrument	Number & validity		Accuracy class
1	Measurement of d.c. insulation resistance windings-to-earth and between windings	Insulation resistance meter JK5053	580576	2022-06-09	/
2	Check of core and frame insulation for liquid-immersed transformers with core or frame insulation				
3	Measurement of dissipation factor (tanδ) of the insulation system capacitances	Full-automatic frequency conversion anti-jamming dielectric dissipation tester JYC	02160521	2022-06-09	/
4	Determination of capacitances windings-to-earth and between windings				
5	Measurement of bushing capacitances and dielectric dissipation factor (tanδ)				
6	Check the ratio and polarity of built-in current transformers	Transformer comprehensive tester JYH-III	1723345	2022-06-10	/
7	Insulation test of auxiliary wiring (AuxW)	AC withstand voltage test system JTKZ	742-152	2022-06-09	/
8	Measurement of winding resistance	DC resistance tester JYR40E	01157965	2022-06-09	/
9	Measurement of voltage ratio and check of phase displacement	Transformer ratio tester JYT-A	04151834	2022-05-09	/
10	Measurement of short-circuit impedance and load loss	Power analyzer WT3000	91N510744	2022-06-09	/
11	Measurement of no-load loss and current				
12	Measurement of no-load loss and current at 90% and 110% of rated voltage				
13	Tests on on-load tap-changers				
14	Applied voltage test	Power-frequency voltage test system TAWF-1000/300	1512153	2021-11-09	/
15	Line terminal AC withstand test	Power analyzer WT3000	91N510744	2022-06-09	/
16	Induced voltage withstand test and induced voltage test with partial discharge measurement	Power analyzer WT3000	91N510744	2022-06-09	/
		TWPD-2FMulti-channel digital partial discharge comprehensive tester TWPD-2F, eight-channel	2105XY121	2022-05-31	/
17	Insulating liquid test, measurement of dissolved gasses in dielectric liquid from each separate oil compartment except diverter switch compartment	Insulating oil dielectric dissipation and resistivity tester HGJD206	1051115	2022-06-09	/
		Dielectric strength tester for insulating oil JKJQ-3	381703415	2021-10-12	/
		Microscale moisture tester CA-2000	E1MB6317	2022-06-09	/
		Gas chromatogram analyzer 2000B	27461380	2022-06-09	/
		Automatic closed flash point tester	JKBS-3	2022-06-09	

Instruments used in the tests

No	Test items	Name & type of instrument	Number & validity		Accuracy class
18	Measurement of frequency response	Transformer winding deformation tester CS-1B	121028	2022-06-09	/
19	Lightning impulse test	Impulse voltage test system RZF-1800	1512391	2021-11-09	/
20	Measurement of no-load excitation characteristics	Power analyzer WT3000	91N510744	2022-06-09	/
21	Long-duration no-load test	Power analyzer WT3000	91N510744	2022-06-09	/
22	Measurement of zero-sequence impedances on three-phase transformers	Power analyzer WT3000	91N510744	2022-06-09	/
23	Measurement of the harmonics of the no-load current				
24	Temperature-rise test	Power analyzer WT3000	91N510744	2022-06-09	/
		Multichannel temperature inspector DT1000	11100033	2022-05-10	/
		Electronic stopwatch SJ9-1	22725	2021-11-10	/
		DC resistance tester JYR40E	01157965	2022-06-09	/
		Infrared thermal camera BritIR	BI30378Y	2022-03-10	
25	Leak testing with pressure for liquid-immersed transformers	Pressure gauge (0~0.1)Mpa	HC69022277554	2021-09-23	/
26	Determination of sound levels	Power analyzer WT3000	91N510744	2022-06-09	/
		Sound level meter HS5633	09016098	2022-04-24	/
27	Mechanical strength test of tank, Vacuum deflection test on liquid-immersed transformers	Pressure gauge (0~0.1)Mpa	HC69022277554	2021-09-23	/
		Stainless steel ruler (0-300)mm/1mm	\	2022-08-15	
		Digital vacuum gauge (0-1000)mbar/0.01mbar	2011-11	2022-01-07	/
28	Measurement of the power taken by the fan and liquid pump motors	Power analyzer WT3000	91N510744	2022-06-09	

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DECLARATION

1. The report is invalid without special seal for testing and page combining seal on the report;
2. The report is invalid if altered;
3. The report is invalid without signatures of persons for drawing up, proof-reading, reviewing and approval;
4. The report is valid only for the inspected and tested samples.

NOTICE

1. In case there is any objection to this report, please raise it to the laboratory within fifteen days starting from the date of receiving the report. Thank you for your cooperation.
2. In case there is no objection, please take back the samples within one month starting from the date of receiving the report, when the manufacturer is going to take back the samples, certificate for sample taking and along with the written approval for the report should be brought in presence, only then the samples could be taken back. On time due, the samples will be in the laboratory's own disposal.

The test report is in total 48 pages including 21 figures and 1 photo

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